

Ocean reflectance spectra at the red, near-infrared, and shortwave infrared from highly turbid waters: A study in the Bohai Sea, Yellow Sea, and East China Sea

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Abstract

Normalized water-leaving radiance spectra $nL_w(\lambda)$ at the red, near-infrared (NIR), and shortwave infrared (SWIR) are quantified and characterized in highly turbid waters of the western Pacific using 3 yr (2009–2011) observations from the Moderate Resolution Imaging Spectroradiometer on the satellite Aqua. $nL_w(645; \text{red})$, $nL_w(859; \text{NIR})$, and $nL_w(1240; \text{SWIR})$ were higher in the coastal region and river estuaries, with SWIR $nL_w(1240)$ reaching up to $\sim 0.2 \text{ mW cm}^{-2} \mu\text{m}^{-1} \text{ sr}^{-1}$ in Hangzhou Bay during winter. The NIR ocean-reflectance spectral shape represented by the ratio of the normalized water-leaving reflectance $\rho_{wN}(\lambda)$ at the two NIR bands $\rho_{wN}(748) : \rho_{wN}(869)$ is highly dynamic and region-dependent. The NIR spectral feature associated with the sediment source from the Yellow River and Ancient Yellow River is noticeably different from that of the Yangtze River. There are non-negligible SWIR $nL_w(1240)$ contributions for waters with the NIR $nL_w(859) > \sim 2.5 \text{ mW cm}^{-2} \mu\text{m}^{-1} \text{ sr}^{-1}$. Estimation of the NIR ocean reflectance with iterative approaches might only be accurate for turbid waters with $nL_w(859) < \sim 1.5 \text{ mW cm}^{-2} \mu\text{m}^{-1} \text{ sr}^{-1}$. Thus, the SWIR atmospheric correction algorithm for satellite ocean-color data processing is indispensable to derive accurate $nL_w(\lambda)$ for highly turbid waters. Current existing satellite algorithms for chlorophyll *a*, diffuse attenuation coefficient at the wavelength of 490 nm ($K_d(490)$), total suspended matter, and inherent optical properties (IOPs) using $nL_w(\lambda)$ at the red band for coastal waters are limited and can only be applied to turbid waters with $nL_w(859) < \sim 1.5 \text{ mW cm}^{-2} \mu\text{m}^{-1} \text{ sr}^{-1}$. Thus, the NIR $nL_w(\lambda)$ measurements are required to characterize water properties for highly turbid waters. Based on the fact that pure water absorption is significantly larger than other absorption components in the NIR wavelengths, we show that it is feasible to analytically derive accurate IOP data for turbid waters with combined satellite-measured visible-NIR $nL_w(\lambda)$ spectra data.

Since the launch of the Coast Zone Color Scanner in 1978 (Gordon et al. 1980; Hovis et al. 1980), satellite ocean-color remote sensing has been mostly focusing on the visible wavelengths (i.e., ocean color). A variety of satellite ocean-color algorithms that use normalized water-leaving radiance spectra $nL_w(\lambda)$ in the blue and green wavelengths for retrieval of ocean parameters such as chlorophyll *a* (Chl *a*), water diffuse attenuation coefficient at the wavelength of 490 nm $K_d(490)$, colored dissolved organic matter (CDOM), calcium particulate inorganic carbon (PIC), and ocean's inherent optical properties (IOPs) have been developed, evaluated, and validated. High-quality ocean-color products with uncertainties for $nL_w(\lambda)$ at the blue within $\sim 5\%$ and Chl *a* within $\sim 35\%$ have been produced in global open oceans (McClain 2009). These satellite ocean-color products derived from the visible wavelengths have been helping us to study and understand the global and regional ocean optical, biological, and ecological processes and phenomena, as well as for monitoring the natural hazards. These include ocean's global-scale biological and biogeochemical variability, ocean response to a short-term weather event, effects of volcano eruption on ocean environmental changes, meso-scale ocean processes, phytoplankton blooms in the open oceans, sea surface temperature (SST) variability and ocean circulations, and harmful algae blooms.

Even though satellite ocean-color data have been widely used in open oceans, there are still many challenges such as atmospheric correction (IOCCG 2010), and IOP complexity in the coastal waters (especially turbid waters), which leads to biased ocean optical retrievals and consequently poor accuracy of the ocean-color products. Wang et al. (2012b) show that normalized water-leaving radiance derived with the near-infrared (NIR) atmospheric correction can be significantly lower, and even become negative in the blue bands in the highly turbid Yellow Sea region. In the Chesapeake Bay region, Chl *a* values from the Moderate Resolution Imaging Spectroradiometer (MODIS) standard product (version R2009) are overestimated by $\sim 59\%$ in the lower Bay and $\sim 137\%$ for the turbid upper Bay region (Son and Wang 2012). These challenges significantly limit the applications of satellite ocean-color data in coastal regions. For the open ocean, there is usually little variation of $nL_w(\lambda)$ at wavelengths of the red, NIR, and shortwave infrared (SWIR). The ocean is generally black in the NIR (certainly at the SWIR) wavelengths for the open ocean. Thus, ocean-color products from satellite ocean-color sensors such as MODIS on the satellite Aqua have been routinely derived using the two MODIS NIR bands (748 and 869 nm; Gordon and Wang 1994), with the assumption of a black ocean in the NIR for the open ocean. For turbid waters in coastal regions, however, the NIR black-ocean assumption is often invalid (Ruddick et al. 2000; Siegel et al. 2000; Wang and Shi 2005), leading to errors in the satellite-derived ocean-color

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products (Wang 2007; Wang et al. 2009b). Some modifications have been implemented to account for the NIR ocean contributions for productive (but not very turbid) nearshore or coastal waters (Stumpf et al. 2003; Bailey et al. 2010). In effect, various investigators have tried to remove the NIR $nL_w(\lambda)$ contributions from the top-of-atmosphere (TOA) radiances, so that a 'black pixel' could be provided for the Gordon and Wang (1994) type atmospheric correction algorithm. It has been reported that significant ocean contributions can also be found at the MODIS SWIR 1240 nm band in regions with highly turbid waters, such as Hangzhou Bay in China, La Plata River Estuary in South America (Shi and Wang 2009a, 2010a), and China's Lake Taihu (Wang et al. 2011, 2013b). This is attributed to the high backscattering in the SWIR 1240 nm by suspended particles, which is not negligible in comparison to the light absorption by pure water at the wavelength of 1240 nm. Even though significant $nL_w(\lambda)$ in the red, NIR, and SWIR bands can indeed pose challenges for atmospheric correction of satellite ocean-color data in turbid ocean and inland freshwaters, it also provides further optical information about the ocean (water) beyond traditional blue and green wavelengths to address the complexity of coastal and inland turbid waters. Recently, the corresponding ocean-color retrieval algorithms focusing on the red and NIR wavelengths in the coastal regions are getting much more attention. Indeed, satellite and in situ $nL_w(\lambda)$ data in the red and NIR wavelengths have been used to characterize and quantify water properties in coastal and inland waters such as Chl *a* concentration (Gitelson et al. 2007), floating green algae blooms (Shi and Wang 2009b), river plumes (Shi and Wang 2009c), total suspended matter (TSM [Shen et al. 2010; Zhang et al. 2010; Son and Wang 2012]), the light diffuse attenuation coefficient (Wang et al. 2009a; Zhang et al. 2012), ocean optical and biological property variations in the Korean dump site of the Yellow Sea (Son et al. 2011), and other changes of coastal environments as well as the inland freshwater environment (Wang et al. 2011, 2012a, 2013b).

Our knowledge of the optical features in the red, NIR, and SWIR wavelengths for the coastal waters is still limited (particularly for optical information at the NIR and SWIR bands), although the optical information in the red, NIR, and SWIR can be used to derive a variety of ocean (water) products in coastal and inland waters to address the dynamic coastal ocean and inland water environments. Wang and Shi (2005) first quantitatively evaluated the NIR $nL_w(\lambda)$ contributions over the U.S. east coastal region using MODIS measurements, and Wang (2007) proposed the SWIR-based atmospheric correction algorithms for turbid coastal and inland waters. In recent years, satellite-derived NIR $nL_w(\lambda)$, as well as some examples of the SWIR $nL_w(\lambda)$, have been successfully used (Shi and Wang 2012a; Wang et al. 2012a). On the other hand, Ruddick et al. (2006) showed that the shape of the NIR $nL_w(\lambda)$ spectrum is determined largely by pure water absorptions. Thus, the ratio of $nL_w(\lambda)$ at the two NIR bands (e.g., 748 and 869 nm for MODIS) is almost a constant (similarity spectrum at the NIR bands; Ruddick et al. 2006). However, Doron et al. (2011) found

that, based on the in situ data, model simulations, and satellite remote-sensing data, the variation of spectral shape in terms of $nL_w(\lambda)$ ratio at the two NIR bands is quite significant, and NIR similarity spectrum can only be valid in moderately turbid waters.

In a recent study, Shi and Wang (2009a) showed that the ocean black pixel assumption for the SWIR 1640 and 2130 nm bands is generally valid for the world's most turbid waters. Thus, satellite measurements at these two MODIS SWIR bands can be used to derive $nL_w(\lambda)$ spectra from the visible to the NIR and SWIR 1240 nm bands (Wang 2007). In this study, MODIS-Aqua measurements at the SWIR 1640 and 2130 nm bands are used to carry out atmospheric correction in order to derive $nL_w(\lambda)$ in the visible, NIR, and SWIR 1240 nm bands. We use highly turbid waters of the Bohai Sea (BS), Yellow Sea (YS), and East China Sea (ECS) to characterize and quantify $nL_w(\lambda)$ spectral features in wavelengths of the red (645 nm), NIR (748, 859, and 869 nm), and SWIR (1240 nm), and show some important implications and applications for satellite ocean- (water) color remote sensing in turbid coastal regions and inland fresh lake waters. Three major issues and concerns are addressed: (1) spectral features and seasonal variability of $nL_w(\lambda)$ in the red, NIR, and SWIR bands in highly turbid waters; (2) spectral shape of $nL_w(\lambda)$ in the NIR wavelengths for different regions; and (3) implications and applications of the spectral reflectance features in the red, NIR, and SWIR wavelengths for atmospheric correction for satellite ocean-color remote sensing, IOP algorithm developments in coastal and inland turbid waters, and retrievals of biological and biogeochemical parameters over highly turbid waters.

Methods

The Bohai Sea, Yellow Sea, and East China Sea—The BS, YS, and ECS are the three major marginal seas in the western Pacific Ocean bounded by China, Korea, and Japan (Fig. 1). They cover some of the most turbid waters in the world (Shi and Wang 2010a,b, 2012a). Major rivers in this region, such as the Yangtze River and Yellow River, transport large amounts of sediments into the BS, YS, and ECS (Milliman and Meade 1983). High loading of sediment concentration (Shi and Wang 2012a; Wang et al. 2007) in the continental shelf regions of these three seas is a significant feature. TSM concentration in the region can reach up to $\sim 1000 \text{ g m}^{-3}$ (Zhang et al. 2010). CDOM concentrations in the BS, YS, and ECS are highly correlated to the river plume and ocean salinity (Gong 2004; Bai et al. 2013). CDOM absorption coefficient at 400 nm, $a_{\text{CDOM}}(400)$, can range from near 0 m^{-1} in the open ocean to $> 1.0 \text{ m}^{-1}$ in the Yangtze River Estuary (Bai et al. 2013).

Seasonal monsoons, the Kuroshio Current, and semidiurnal and diurnal tides (Fang 1994) are primary processes that dominate the ocean hydrography, ocean circulations, SST, ocean stratification, and ocean fronts in these three seas. Images of MODIS Chl *a* concentration show significant cross-shelf penetrating fronts off the southeast

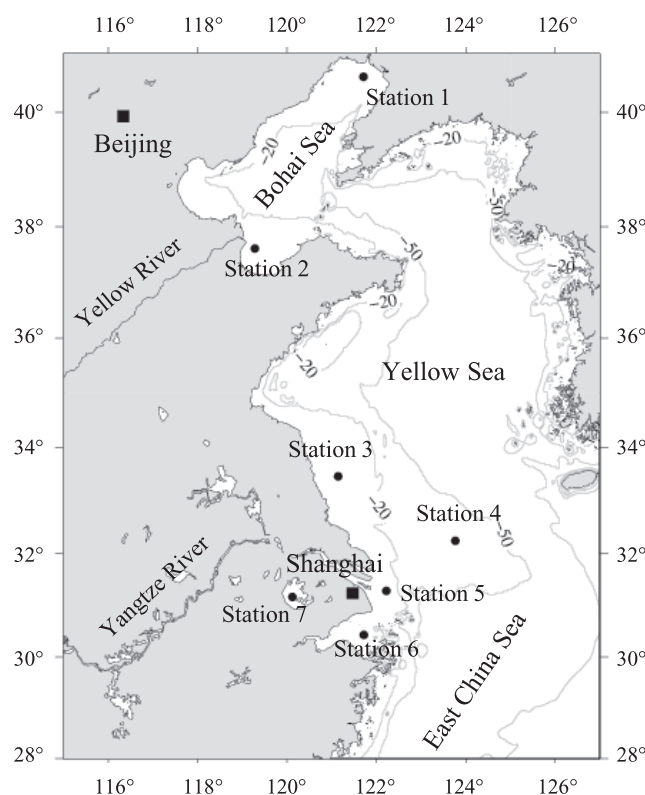


Fig. 1. Map of the Bohai Sea, Yellow Sea, and East China Sea. Locations of the seven pseudo-stations for representative turbid waters in the three seas are also marked. Sta. 1 (40.60°N, 121.71°E) and 2 (37.61°N, 119.18°E) are located in the Liaohe River Estuary and Yellow River Estuary of the Bohai Sea, respectively. Sta. 3 (33.45°N, 121.14°E) is located in the Subei Shoal of the Yellow Sea. Sta. 4 (32.23°N, 123.77°E), 5 (31.27°N, 122.23°E), 6 (30.43°N, 121.72°E), and 7 (31.16°N, 120.12°E) are chosen to quantify the optical properties in the central East China Sea plume, Yangtze River Estuary, Hangzhou Bay, and Lake Taihu, respectively.

coast of China. Frequent occurrences of green algae blooms in the YS during the spring and early summer significantly affect the ecosystem of this region. Existence of the ocean bottom sand ridges in the BS also leads to enhanced TSM and $K_d(490)$ in the eastern BS (Shi and Wang 2012a). The effects of spring-neap tide in this region on the TSM and $K_d(490)$ are found to be as significant as the seasonal variability (Shi et al. 2011).

High sediment loadings in the BS, YS, and ECS and the complexities of the physical, biological, and biogeochemical processes mean that the optical features of this region are also complicated. Strong seasonal $nL_w(\lambda)$ variations in the green (555 nm), red (645 nm), and NIR (859 nm) bands are found in coastal highly turbid waters, while a seasonal $nL_w(\lambda)$ change in the blue band is relatively small (Shi and Wang 2012a). In the coastal region of the BS (Sta. 1, 2 in Fig. 1), Subei Shoal of the YS (Sta. 3 in Fig. 1), Yangtze River Estuary (Sta. 5 in Fig. 1), Hangzhou Bay (Sta. 6 in Fig. 1), and Lake Taihu (Sta. 7 in Fig. 1), normalized water-leaving radiance at the red band $nL_w(645)$ can be over $\sim 5 \text{ mW cm}^{-2} \mu\text{m}^{-1} \text{ sr}^{-1}$ in the winter season (Shi and Wang 2012a; Wang et al. 2011). Normalized water-leaving

radiance at 859 nm ($nL_w(859)$) normally is also over $\sim 2 \text{ mW cm}^{-2} \mu\text{m}^{-1} \text{ sr}^{-1}$ at these stations. For the central BS, YS, and ECS, $nL_w(\lambda)$ spectra are significantly different from the coastal regions.

Atmospheric correction using the MODIS SWIR 1640 and 2130 nm bands—Similar to the standard NIR atmospheric correction algorithm (Gordon and Wang 1994; IOCCG 2010) with two NIR bands (748 and 869 nm for MODIS), the SWIR-based atmospheric correction algorithm with MODIS SWIR 1240 and 2130 nm bands or 1640 and 2130 nm bands (Wang 2007) has been used in deriving the normalized water-leaving (or surface-leaving) reflectance spectra $\rho_{wN}(\lambda)$, where $\rho_{wN}(\lambda) = \pi nL_w(\lambda)/F_0(\lambda)$ with $F_0(\lambda)$ the extraterrestrial solar irradiance. It is noted that in this paper both $nL_w(\lambda)$ and $\rho_{wN}(\lambda)$ are used and they differed by a constant $\pi/F_0(\lambda)$. Definitions of $nL_w(\lambda)$ and $\rho_{wN}(\lambda)$ can be found in various references (Morel and Gentili 1996; Gordon 2005; Wang 2006), and furthermore remote-sensing reflectance $R_{rs}(\lambda) = nL_w(\lambda)/F_0(\lambda)$. The SWIR atmospheric correction algorithm is based on the fact that waters are strongly absorbing in the SWIR wavelengths rather than those in the NIR wavelengths (Hale and Querry 1973). Thus, the black pixel assumption is still valid at the SWIR bands for turbid waters while this assumption fails at the NIR bands. Consequently, MODIS measurements at the SWIR bands can be used to determine the aerosol type and estimate the atmospheric effects in the visible and NIR by extrapolating the aerosol effect from the SWIR wavelengths to visible and NIR wavelengths (Wang 2007).

However, because 4 out of 10 detectors for the MODIS-Aqua SWIR 1640 nm band is inoperable (<http://mcst.gsfc.nasa.gov/calibration/known-issues>), the SWIR band set of 1240 and 2130 nm is normally used to carry out atmospheric correction. Even though the ocean radiance contribution for the MODIS 1240 nm is generally negligible in turbid waters such as regions along the U.S. east coast (Wang and Shi 2007; Son and Wang 2012), some slight ocean radiance contributions at 1240 nm are observed in extremely turbid waters (e.g., Hangzhou Bay in China's east coast, La Plata River Estuary in South America (Shi and Wang 2009a), and China's Lake Taihu; Wang et al. 2011, 2013b). Actually, the MODIS SWIR band set of 1640 and 2130 nm has been used to derive water optical properties and show reasonable accuracy in the satellite-derived $nL_w(\lambda)$ spectra in highly turbid inland waters (Wang et al. 2013b) and over the sea ice in the BS (Shi and Wang 2012b,c). In this study, to characterize and quantify the optical properties in the red, NIR, and SWIR (1240 nm) bands, MODIS-Aqua measurements with the band set of 1640 and 2130 nm are used to carry out atmospheric correction for deriving $nL_w(\lambda)$ at wavelengths of 645, 748, 859, 869, and 1240 nm.

Part of the BS region is frequently covered with sea ice during the winter season (Shi and Wang 2012b,c). In this study, a regional sea ice detection algorithm (Shi and Wang 2012b) is applied to detect sea ice coverage in the BS and YS. Sea ice-covered and contaminated pixels are then excluded in quantifications and evaluations of the reflectance spectral features for turbid waters in the region.

Data and data analysis method—Three-year MODIS-Aqua observations from 2009 to 2011 are used to derive $nL_w(645)$, $nL_w(748)$, $nL_w(859)$, $nL_w(869)$, and $nL_w(1240)$ using the SWIR atmospheric correction with the band set of 1640 and 2130 nm (Wang 2007). It is noted that MODIS ocean-color bands at 667 and 678 nm are normally saturated in turbid waters such as the BS, YS, and ECS because of strong ocean signal in the red wavelengths (limitation from sensor design). Thus, $nL_w(\lambda)$ values at these two bands are not used in this study. The sensor at the wavelengths of 748 and 869 nm can also become saturated for cases with extremely turbid waters when $nL_w(748)$ and $nL_w(869)$ exceed ~ 4.0 and ~ 2.4 $\text{mW cm}^{-2} \mu\text{m}^{-1} \text{sr}^{-1}$, respectively; therefore, $nL_w(645)$, $nL_w(859)$, and $nL_w(1240)$ are used instead to evaluate the spectral features and their seasonal changes in the red, NIR, and SWIR wavelengths for highly turbid waters. However, for not extremely turbid waters, MODIS-Aqua $nL_w(748)$ and $nL_w(869)$ data can still be converted to $\rho_{wN}(748)$ and $\rho_{wN}(869)$, and then used to examine the spectral shape of the NIR $\rho_{wN}(\lambda)$ represented by the ratio of $\rho_{wN}(\lambda)$ between the two NIR bands. Following Lee et al. (1999), the subsurface remote-sensing reflectance $r_{rs}(\lambda)$ in sr^{-1} can be calculated as from remote-sensing reflectance $R_{rs}(\lambda)$,

$$r_{rs}(\lambda) = \frac{R_{rs}(\lambda)}{0.52 + 1.7R_{rs}(\lambda)} \quad (1)$$

In this study, the spectral features in the red, NIR, and SWIR are evaluated with $nL_w(\lambda)$. The spectral shape at the NIR wavelengths and ‘similarity spectrum’ in the NIR wavelengths are examined in terms of the reflectance ratio between $\rho_{wN}(748)$ and $\rho_{wN}(869)$ and some IOP parameters are derived from $r_{rs}(\lambda)$, and their implications are evaluated and discussed. In addition, to assess the spectral shape of the NIR reflectance in highly turbid waters, $\rho_{wN}(\lambda)$ data at the two MODIS NIR bands are required. Seven pseudostations (Fig. 1) were chosen to quantify and characterize the NIR spectral shapes of turbid waters in the three seas as well as Lake Taihu. Sta. 1 is located in the Liaohe River Estuary of the northern BS. Sta. 2 is in the turbid waters around the Yellow River Estuary. Sta. 3–7 are at the Subei Shoal, central ECS plume, Yangtze River Estuary, Hangzhou Bay, and Lake Taihu, respectively.

Spectral features of $nL_w(\lambda)$ in the red, NIR, and SWIR bands

$nL_w(\lambda)$ in the red, NIR, and SWIR bands—Three-year composite images of $nL_w(645)$ and $nL_w(859)$ (not shown here) for the period between 2009 and 2011 have been examined, and are similar to the climatology $nL_w(645)$ and $nL_w(859)$ from 2002 to 2009 (Shi and Wang 2012a). Significantly enhanced $nL_w(645)$ is located at the Subei Shoal of the YS with values over ~ 6 $\text{mW cm}^{-2} \mu\text{m}^{-1} \text{sr}^{-1}$. In the Yangtze River Estuary and Hangzhou Bay, values in $nL_w(645)$ are ~ 5 $\text{mW cm}^{-2} \mu\text{m}^{-1} \text{sr}^{-1}$, which are a little bit lower than that at the Subei Shoal in the YS. Actually, the NIR $nL_w(859)$ can reach over ~ 3 $\text{mW cm}^{-2} \mu\text{m}^{-1} \text{sr}^{-1}$ for a large portion of Hangzhou Bay. In fact, the pattern of

enhanced red $nL_w(645)$ in the coastal regions of the BS, YS, and ECS is similar to the climatology of $nL_w(443)$ (blue), $nL_w(555)$ (green), $K_d(490)$, and TSM concentration (Shi and Wang 2012a).

To further characterize and quantify the seasonal variations of the spectral properties in the BS, YS, and ECS, composites of $nL_w(645)$, $nL_w(859)$, and $nL_w(1240)$ in the four seasons for the period between 2009 and 2011 are generated (Fig. 2). For the four different seasons, cloud occurs more frequently during the summer season. Thus, cloud-free pixels for generating seasonal composites may be different in Fig. 2. However, $nL_w(\lambda)$ spectra at these seven stations are typical for turbid waters with enhanced values in the red and NIR bands, and are consistent with the previous study (Shi and Wang 2012a). Similar to $nL_w(443)$ for blue band and $nL_w(555)$ for green band (Shi and Wang 2012a), significant seasonal variability of $nL_w(645)$, $nL_w(859)$, and $nL_w(1240)$ in terms of spatial patterns and $nL_w(\lambda)$ magnitudes can be found in the three seas. The regional and seasonal differences of $nL_w(\lambda)$ at the red, NIR, and SWIR wavelengths can be attributed to the different particle concentrations for different regions and seasons (Shi and Wang 2012a). The major driving forcings for the seasonal variability in the BS, YS, and ECS are seasonal winds, ocean physical processes (e.g., wave, turbulence, mixing), and river discharge, as well as coastal bathymetry (Shi and Wang 2012a). Of the four seasons, summer is the least turbid season. This is reflected with significantly weakened $nL_w(645)$ (Fig. 2b), $nL_w(859)$ (Fig. 2f), and near-zero $nL_w(1240)$ (Fig. 2j) in highly turbid regions. In Hangzhou Bay, $nL_w(645)$, $nL_w(859)$, and $nL_w(1240)$ in the summer are ~ 4 – 5 , ~ 1 – 2 , and ~ 0 – 0.02 $\text{mW cm}^{-2} \mu\text{m}^{-1} \text{sr}^{-1}$, respectively, compared with the corresponding values of ~ 6 – 7 , ~ 2 – 3 , and ~ 0.06 – 0.12 $\text{mW cm}^{-2} \mu\text{m}^{-1} \text{sr}^{-1}$ for the 3 yr mean values. Similarly, significant drops of $nL_w(645)$, $nL_w(859)$, and $nL_w(1240)$ can be found in highly turbid waters of the BS.

In the winter, however, significantly enhanced $nL_w(645)$ (Fig. 2d), $nL_w(859)$ (Fig. 2h), and $nL_w(1240)$ (Fig. 2l) can be found in coastal regions. In the Subei Shoal, $nL_w(645)$ reaches over ~ 9 $\text{mW cm}^{-2} \mu\text{m}^{-1} \text{sr}^{-1}$, and is notably higher than that in Hangzhou Bay. It is worth noting that $nL_w(645)$ in the central ECS also becomes significant because of the sediment plumes in that region (Shi and Wang 2010b). The NIR $nL_w(859)$ along the coastal region shows a broad increase similar to $nL_w(645)$ in comparison with other seasons. However, $nL_w(859)$ is generally higher in Hangzhou Bay than that in the Subei Shoal region. This is remarkably different from the $nL_w(645)$ map in Fig. 2d, which shows that $nL_w(645)$ in the Subei Shoal is generally higher than that in Hangzhou Bay.

Figure 2l provides large-scale satellite retrievals of $nL_w(\lambda)$ in the SWIR 1240 nm band, and clearly demonstrates that highly turbid waters such as Hangzhou Bay indeed have radiance contributions at the SWIR 1240 nm band. Non-negligible $nL_w(1240)$ can also be identified in the broad coastal regions of the Subei Shoal and Yangtze River Estuary. Similar to the pattern of $nL_w(859)$ (Fig. 2h), $nL_w(1240)$ in Hangzhou Bay is more enhanced than those in other areas such as the Subei Shoal of the YS. Values in

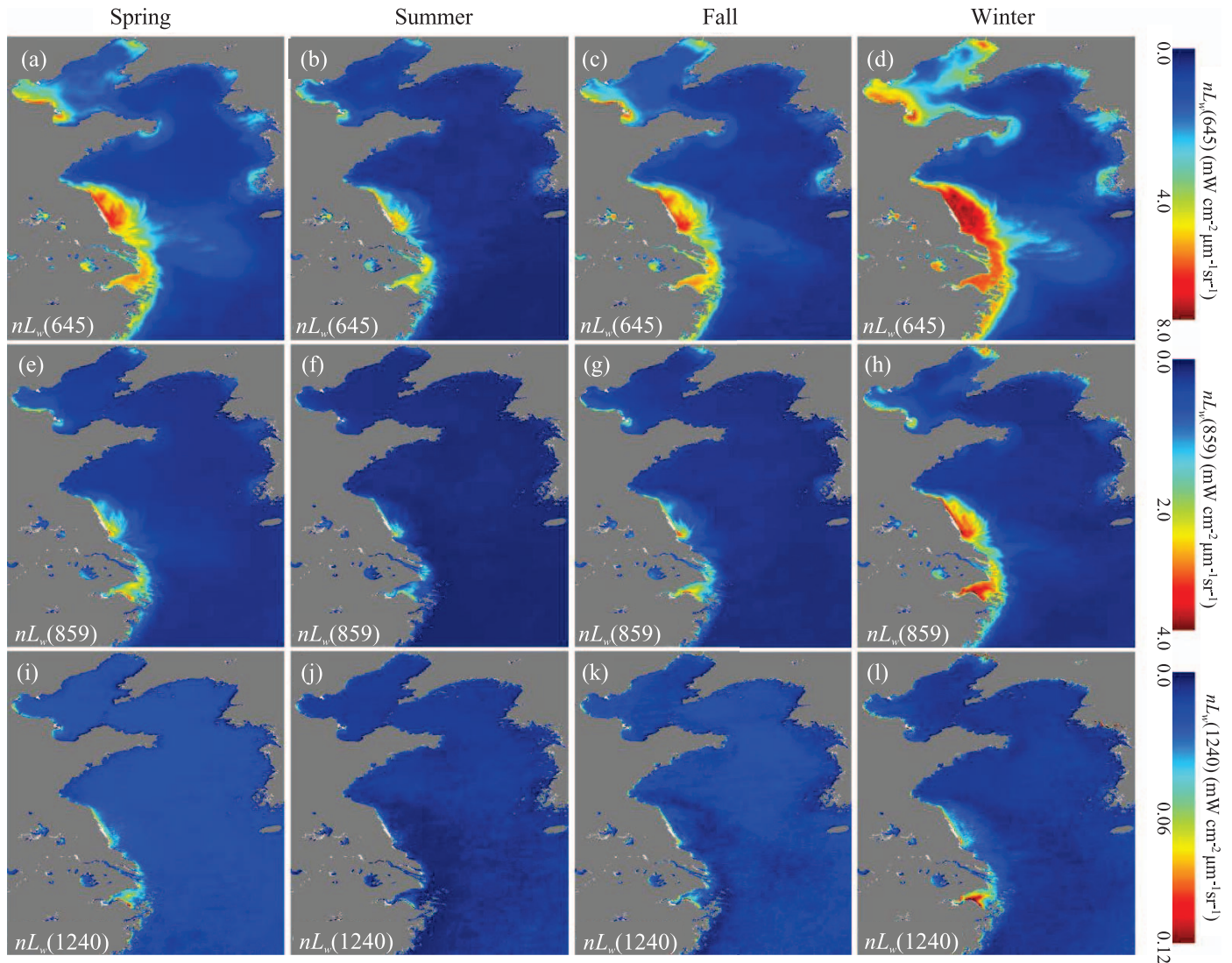


Fig. 2. Three-year mean seasonal images of (a–d) $nL_w(645)$, (e–h) $nL_w(859)$, and (i–l) $nL_w(1240)$ derived from MODIS-Aqua observations between 2009 and 2011 in the Bohai Sea, Yellow Sea, and East China Sea for the season of (a, e, i) spring (March–May), (b, f, j) summer (June–August), (c, g, k) autumn (September–November), and (d, h, l) winter (December–February).

$nL_w(1240)$ are actually significant in the central Hangzhou Bay. In fact, the mean $nL_w(1240)$ during the winter season between 2009 and 2011 was over $\sim 0.15 \text{ mW cm}^{-2} \mu\text{m}^{-1} \text{ sr}^{-1}$ in part of Hangzhou Bay. For some winter days, $nL_w(1240)$ even reached $\sim 0.30 \text{ mW cm}^{-2} \mu\text{m}^{-1} \text{ sr}^{-1}$. It is noted that contribution of $nL_w(1240)$ with a value of $\sim 0.1 \text{ mW cm}^{-2} \mu\text{m}^{-1} \text{ sr}^{-1}$ is very significant. In fact, the TOA radiance at the SWIR 1240 nm for a typical clear sky over ocean is generally $\sim 0.1 \text{ mW cm}^{-2} \mu\text{m}^{-1} \text{ sr}^{-1}$.

For the spring and autumn seasons, enhanced $nL_w(645)$; Fig. 2a, c) can be found in the coastal regions of the three seas. The magnitudes of $nL_w(645)$ and $nL_w(859)$, however, are lower than those in the winter season, but still significantly higher than those in the summer season. During the autumn, $nL_w(1240)$ in most of the coastal regions is much smaller except for some moderate values $\sim 0.05 \text{ mW cm}^{-2} \mu\text{m}^{-1} \text{ sr}^{-1}$ in part of Hangzhou Bay (Fig. 2k). In the spring season, non-negligible $nL_w(1240)$;

Fig. 2i) can be found in the broad coastal regions of the YS and ECS, but values of $nL_w(1240)$ are notably lower than those in the winter season. It should be noted that some very low $nL_w(1240)$ values may not be meaningful because of sensor noise error in $nL_w(1240)$ data ($\sim 1.82 \times 10^{-3} \text{ mW cm}^{-2} \mu\text{m}^{-1} \text{ sr}^{-1}$ [Xiong et al. 2010; Wang and Shi 2012]) and uncertainty from atmospheric correction.

Comparison of composite maps of $nL_w(645)$ and $nL_w(859)$ shows that $nL_w(859)$ values in Hangzhou Bay are, in general, larger than those in the Subei Shoal region, although $nL_w(645)$ values in Hangzhou Bay are notably smaller than those in the Subei Shoal region. It is also interesting to note that the sediment source in the Subei Shoal is from the Ancient Yellow River, while the southbound Yangtze River plume is responsible for large amounts of sediments in Hangzhou Bay (Milliman et al. 1985a,b). Sediment particle features are highly related to remotely sensed $nL_w(\lambda)$ spectra in turbid waters (Lu et al.

2013). This indeed indicates that the optical features in the NIR wavelengths can provide valuable information about the particle properties of the sediments, such as particle size, composition, and refractive index.

Highly turbid regions as represented with strong $nL_w(645)$ in the red are mostly black at the SWIR 1240 nm with $nL_w(1240)$ near 0. This is consistent with the evaluation of black pixel assumption in the region (Shi and Wang 2009a), which suggests that atmospheric correction with the SWIR band set of 1240 and 2130 nm is generally valid. However, $nL_w(1240)$ for Hangzhou Bay and a portion of the Subei Shoal is not always negligible. This is especially true for the winter season (Fig. 2l). For some part of Hangzhou Bay, $nL_w(1240)$ can be over $\sim 0.1 \text{ mW cm}^{-2} \mu\text{m}^{-1} \text{ sr}^{-1}$.

Spectral shape of ocean reflectance in the NIR wavelengths—Spectral shapes in the NIR ocean reflectance in terms of the reflectance ratio between $\rho_{wN}(\lambda)$ at two NIR bands are important for satellite ocean-color remote sensing (Doxaran et al. 2002). They are also linked to the IOPs, which reflect the major geophysical and biogeochemical characteristics of the ocean particle constituents, such as particle compositions, size distributions, and particle textures (Babin et al. 2003; Doxaran et al. 2007; Stramski et al. 2007). Ruddick et al. (2006) proposed that a similarity spectrum is valid for waters ranging from moderately to extremely turbid (i.e., the spectral shapes vary little in the turbid to extremely turbid waters). However, Doron et al. (2011) argued that significant spectral variations in the NIR ocean reflectance do indeed exist, and similarity spectrum is only valid in moderately turbid waters.

The NIR ocean-reflectance $\rho_{wN}(\lambda)$ derived using the SWIR atmospheric correction with the band set of 1640 and 2130 nm is used to characterize and quantify the spectral shape of the NIR ocean reflectance as well as its regional variations in the three seas. Except Sta. 4 in the central ECS plume, the NIR ocean-reflectance values in the six stations all have wide $\rho_{wN}(869)$ ranges from ~ 0.01 to ~ 0.10 between 2009 and 2011. Most of the variability can be attributed to the seasonal variability as shown in Fig. 2, as well as other ocean physical, optical, biological, and biogeochemical changes. At Sta. 3 and 5–7, the sensor at the NIR 748 and 869 nm bands are actually saturated because of the strong NIR $\rho_{wN}(\lambda)$ contributions for some MODIS-Aqua granules (normally for cases with $\rho_{wN}(869)$ over ~ 0.09). Thus, these data are not included in the scatter plots. For example, $\rho_{wN}(859)$, which is very close to $\rho_{wN}(869)$, can be up to ~ 0.16 for some observations in Hangzhou Bay.

Figure 3 shows that $\rho_{wN}(748)$ and $\rho_{wN}(869)$ are not linearly correlated for $\rho_{wN}(869)$ ranging from 0 to 0.10. The curvature of $\rho_{wN}(748)$ vs. $\rho_{wN}(869)$ in Fig. 3 suggests that reflectance ratio values in $\rho_{wN}(748):\rho_{wN}(869)$ are not constant and trend lower with $\rho_{wN}(869)$ increasing from 0 to 0.10. However, a quasi-linear relationship between $\rho_{wN}(748)$ and $\rho_{wN}(869)$ can be found for $\rho_{wN}(869)$ ranging from ~ 0 to 0.02 at some stations, such as Sta. 1 and 2 in the BS, Sta. 3 in the YS, and Sta. 6 in Hangzhou Bay. On

the other hand, the trend of $\rho_{wN}(748)$ vs. $\rho_{wN}(869)$ also shows noticeable regional difference (e.g., it shows some differences for waters with different sediment sources). At Sta. 3 (Fig. 3c), $\rho_{wN}(748)$ is ~ 0.10 , which corresponds to $\rho_{wN}(869)$ of ~ 0.07 . In comparison, $\rho_{wN}(748)$ is ~ 0.09 when $\rho_{wN}(869)$ is 0.07 at Sta. 6 in Hangzhou Bay (Fig. 3f). Regional difference of the relationship in $\rho_{wN}(748)$ vs. $\rho_{wN}(869)$ for different stations is further highlighted with wide scattering in Fig. 3h, especially for high $\rho_{wN}(869)$ values. It is noted that values of $\rho_{wN}(748)$ and $\rho_{wN}(869)$ are generally in the low range of each match-up scatter plot during the summer, while match-ups of large $\rho_{wN}(748)$ and $\rho_{wN}(869)$ occur in the winter season.

Sediment sources for the seven stations are not the same. Sediment sources for Sta. 2 and 3 are from Yellow River and Ancient Yellow River, while sediments at Sta. 5–7 are mostly from Yangtze River (Milliman et al. 1985a,b). Further evaluation of the relationship in $\rho_{wN}(748)$ vs. $\rho_{wN}(869)$ shows that the curve trends for Sta. 2 in the Yellow River Estuary and Sta. 3 in the Subei Shoal are similar, while trends for Sta. 5–7 are almost the same. Although $\rho_{wN}(748):\rho_{wN}(869)$ shows big uncertainty for less turbid waters with $\rho_{wN}(869) < \sim 0.02$ (as expected) because of low $\rho_{wN}(869)$ values, small signal:noise-ratio (SNR) values for the two SWIR bands, and uncertainty from atmospheric correction, the NIR reflectance relationships for Sta. 2 and 3 are noticeably different from those of Sta. 5–7 for relatively large $\rho_{wN}(869)$ values as shown in Fig. 3. This suggests that the suspended sediment particle properties such as size distributions, compositions, and textures might be different for the Yellow River and Yangtze River. In fact, a recent study from Lake Taihu shows that particle size distribution plays an important role for the measured $\rho_{wN}(\lambda)$ spectra (Lu et al. 2013). Zhang et al. (2008) also reported that the compositions of river sediments from Yellow River and Yangtze River are different. The regional difference in the NIR spectral features as observed in this study is attributed to the difference of the sediment particle properties at these stations.

Figure 4 further quantifies the NIR reflectance spectral shapes in $\rho_{wN}(748):\rho_{wN}(869)$ for these stations. As expected, $\rho_{wN}(748):\rho_{wN}(869)$ is quite noisy for $\rho_{wN}(869)$ less than ~ 0.02 because of atmospheric correction uncertainties and low SNR for the SWIR band set of 1640 and 2130 nm (Wang and Shi 2012). Because $\rho_{wN}(869)$ values are mostly less than ~ 0.03 for Sta. 2 and 4, large noise in reflectance ratio $\rho_{wN}(748):\rho_{wN}(869)$ does not reflect the trend of $\rho_{wN}(748):\rho_{wN}(869)$ with the variation of $\rho_{wN}(869)$ (not shown here). However, as suggested in Fig. 3, $\rho_{wN}(748):\rho_{wN}(869)$ visibly trends low in Sta. 1 (Fig. 4a), Sta. 3 (Fig. 4b), Sta. 5 (Fig. 4c), Sta. 6 (Fig. 4d), and Sta. 7 (Fig. 4e), with increase of $\rho_{wN}(869)$. Specifically, reflectance ratio in $\rho_{wN}(748):\rho_{wN}(869)$ changes from mean value of ~ 1.7 for $\rho_{wN}(869)$ of ~ 0 –0.02 to ~ 1.3 –1.4 for $\rho_{wN}(869)$ of ~ 0.08 at Sta. 3 in the Subei Shoal of the YS. Apparently, downward trend in $\rho_{wN}(748):\rho_{wN}(869)$ can also be found at Sta. 5 in the Yangtze River Estuary, Sta. 6 in Hangzhou Bay, and Sta. 7 in Lake Taihu. It is also noted that the downward slopes for Sta. 5–7 (Fig. 4c–e) are

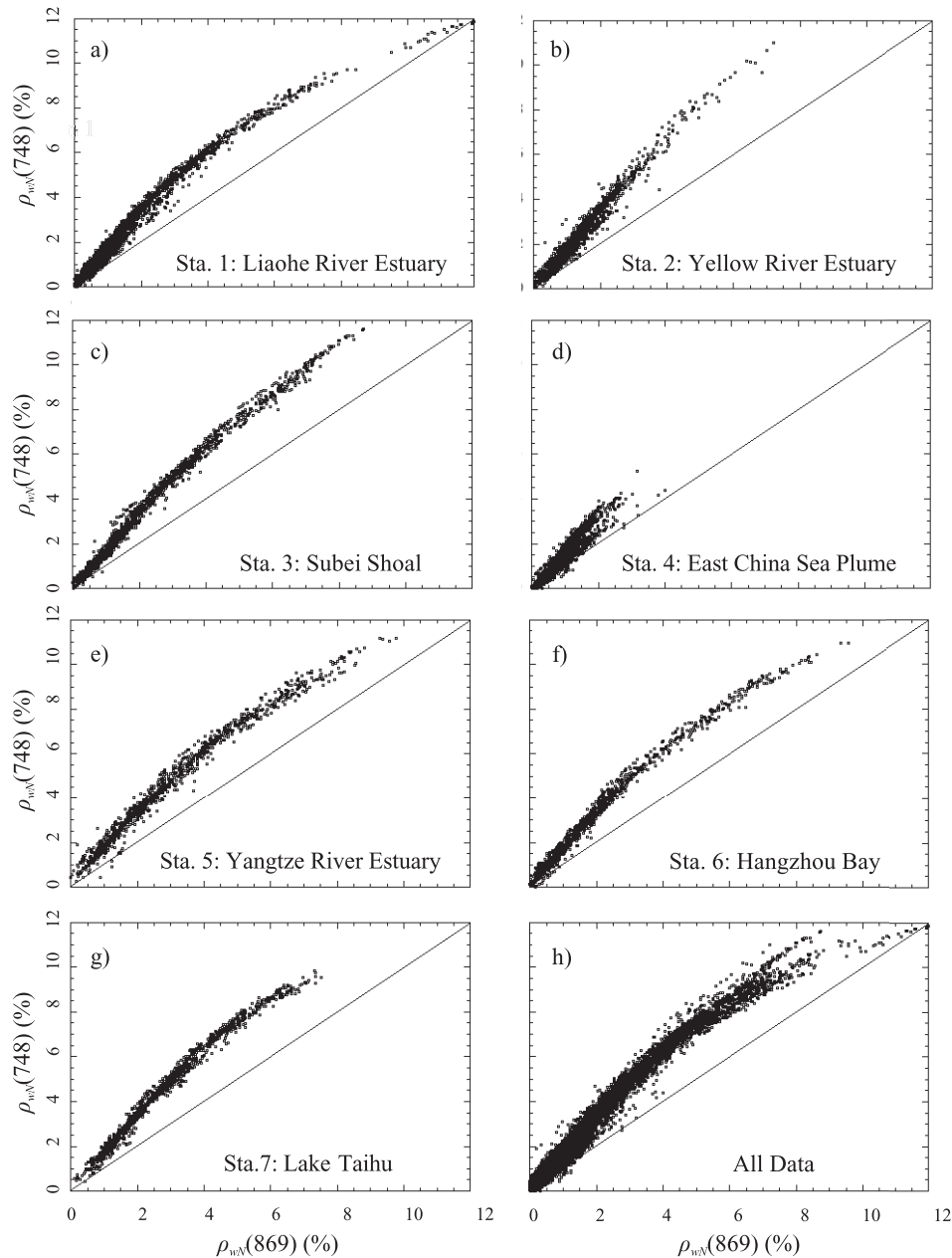


Fig. 3. Scatter plots of MODIS-Aqua-derived $\rho_{wN}(748)$ vs. $\rho_{wN}(869)$ from 2009 to 2011 in a 5×5 box centered at (a–g) Sta. 1–7 and (h) all seven stations.

similar and larger than that at Sta. 3 (Fig. 4b). This reflects the regional difference of trends in $\rho_{wN}(748)$ vs. $\rho_{wN}(869)$ as shown in Fig. 3. For the NIR reflectance spectral shapes from Sta. 1, 3, and 5–7 (Fig. 4f), wide scattering in $\rho_{wN}(748): \rho_{wN}(869)$ as a function of $\rho_{wN}(869)$ mainly represents the variability of the NIR reflectance spectral shapes from different stations.

Some satellite ocean-color remote-sensing implications

The SWIR atmospheric correction algorithm with the SWIR band set of 1640 and 2130 nm allows us to derive a large-scale ocean reflectance in the MODIS NIR and

SWIR (1240 nm) bands over a long time period. The spectral features of ocean reflectance in the red, NIR, and SWIR (1240 nm) bands have important implications and applications for atmospheric correction, IOP retrieval algorithms (e.g., $b_{bp}(\lambda)$, $a(\lambda)$, etc.), and biological and biogeochemical satellite ocean-color algorithms (e.g., Chl a , $K_d(490)$, TSM concentration, etc.) in ocean-color remote sensing over coastal and inland waters.

Implications for atmospheric correction—Although the SWIR atmospheric correction algorithm using the SWIR band set of 1240 and 2130 nm can produce improved $nL_w(\lambda)$ spectra data in turbid regions such as the U.S. east

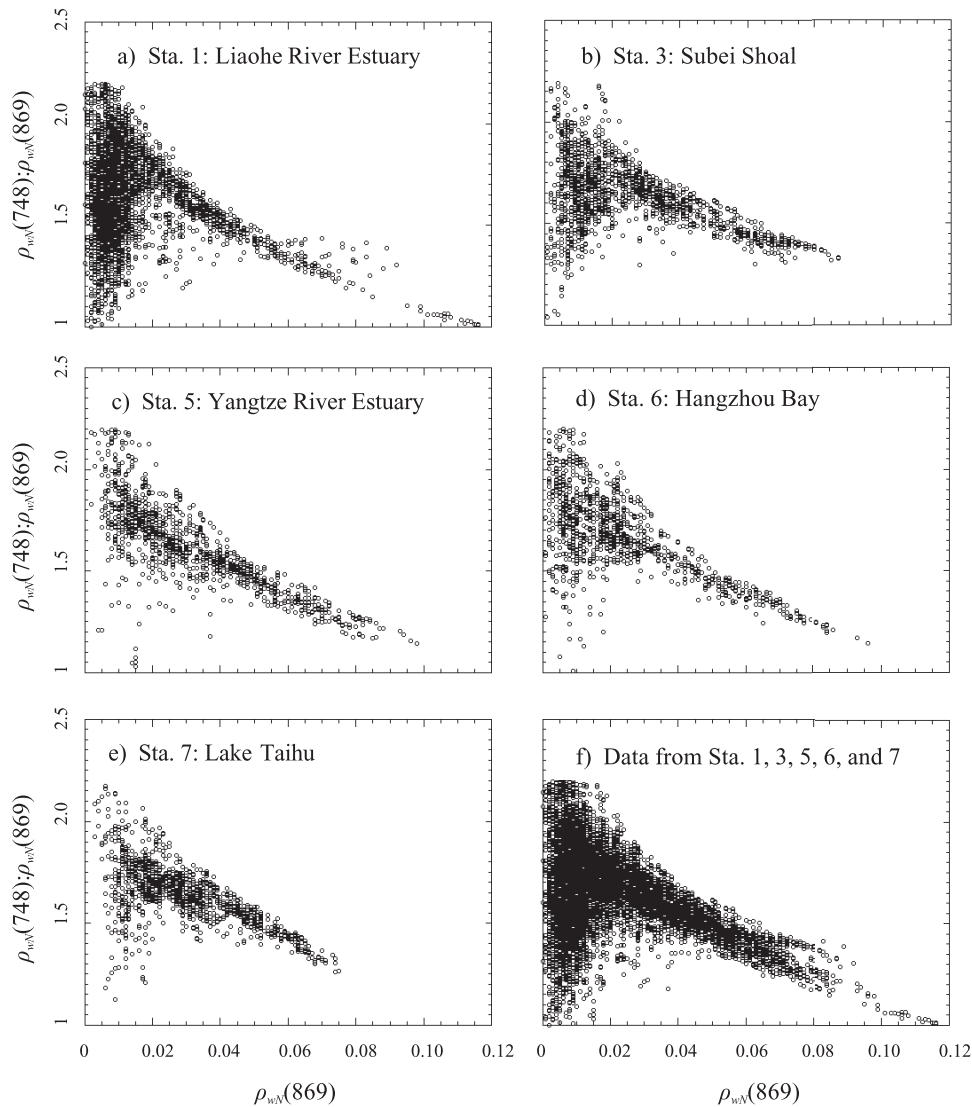


Fig. 4. $\rho_{wN}(748) : \rho_{wN}(869)$ vs. $\rho_{wN}(869)$ between 2009 and 2011 in a 5×5 box centered at (a) Sta. 1, (b) Sta. 3, (c–e) Sta. 5–7, and (f) Sta. 1, 3, 5, 6, and 7.

coastal and China east coastal regions and inland waters (Wang et al. 2007, 2011; Shi and Wang 2012a), it has been found that the MODIS SWIR 1240 nm band is not always black in highly turbid waters such as Hangzhou Bay, La Plata River Estuary, and Lake Taihu (Shi and Wang 2009a; Wang et al. 2011, 2013b). Ocean radiance contributions at the SWIR 1240 nm band result in overcorrection of atmospheric and water surface effects, which lead to errors of MODIS-derived $nL_w(\lambda)$ data at the blue bands reaching $\sim 0.5 \text{ mW cm}^{-2} \mu\text{m}^{-1} \text{ sr}^{-1}$ (Shi and Wang 2009a).

Figure 5 further quantifies the applicability of the SWIR 1240 nm band for the SWIR atmospheric correction algorithm in highly turbid waters. Three-year seasonal maps of $nL_w(1240)$ (Fig. 2i–l) have shown that $nL_w(1240)$ can be significant in Hangzhou Bay. At Sta. 6 in Hangzhou Bay, $\rho_{wN}(1240)$ vs. $\rho_{wN}(859)$ scatter plot shows a generally nonlinear relationship between the ocean-reflectance values at the NIR (859 nm) and the SWIR 1240 nm. Reflectance $\rho_{wN}(1240)$ is flat and close to 0 for $\rho_{wN}(859)$, ranging from

0 to 0.08. This shows that the SWIR 1240 nm band can be assumed to be black for this range; thus, it can be safely used to carry out atmospheric correction for waters with $\rho_{wN}(859)$ of ~ 0 –0.08 (equivalent $nL_w(859) \sim 0$ – $2.5 \text{ mW cm}^{-2} \mu\text{m}^{-1} \text{ sr}^{-1}$). However, $\rho_{wN}(1240)$ can be increased remarkably and becomes non-negligible when $\rho_{wN}(859)$ is over ~ 0.08 . As an example, $\rho_{wN}(1240)$ is ~ 0.01 (equivalent $nL_w(1240) \sim 0.15 \text{ mW cm}^{-2} \mu\text{m}^{-1} \text{ sr}^{-1}$) corresponding to $\rho_{wN}(859)$ of ~ 0.15 . Thus, atmospheric correction with the SWIR band set of 1240 and 2130 nm in the highly turbid waters with $\rho_{wN}(859)$ over ~ 0.08 can lead to errors of $nL_w(\lambda)$ in the visible and NIR wavelengths. It is noted that negative $\rho_{wN}(1240)$ in the scatter plot of Fig. 5 reflects the noise of the SWIR 1240 nm band as well as the uncertainty from the SWIR atmospheric correction.

However, with the derived relationship of $\rho_{wN}(1240)$ vs. $\rho_{wN}(859)$ as shown in Fig. 5, the $\rho_{wN}(1240)$ contribution can be iteratively estimated and removed in atmospheric correction with the SWIR band set of 1240 and 2130 nm.

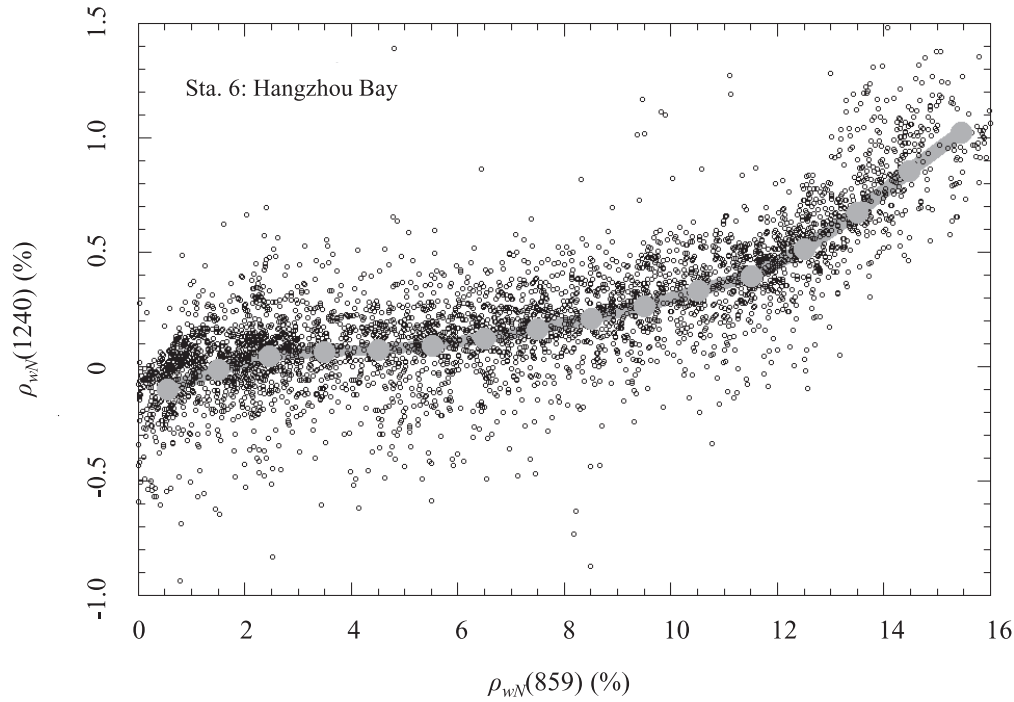


Fig. 5. Scatter plot of MODIS-Aqua-derived $\rho_{wN}(859)$ vs. $\rho_{wN}(1240)$ from 2009 to 2011 in a 5×5 box centered at Sta. 6 in Hangzhou Bay. Large circles represent the running mean of $\rho_{wN}(1240)$ following the change of $\rho_{wN}(859)$. The bin size for mean calculations is 0.005 for $\rho_{wN}(859)$.

This idea has been proposed and tested in a regional atmospheric correction algorithm for an estimation of the NIR $nL_w(\lambda)$ contributions (Wang et al. 2012b, 2013a). There are also other approaches in the estimation of $\rho_{wN}(1240)$ for extremely turbid waters (Wang et al. 2011).

Implications for ocean IOP retrievals—Background for the NIR-based IOP algorithm: Satellite-measured $nL_w(\lambda)$ data are intrinsically related to the constituents of sea waters such as CDOM, phytoplankton particles, inorganic sediments, etc. The specific spectral features of each constituent, such as absorption coefficient ($a(\lambda)$) and backscattering coefficient (e.g., $b_{bp}(\lambda)$), determine the spectral variability of $nL_w(\lambda)$ spectra and remote-sensing reflectance just beneath the surface ($r_{rs}(\lambda)$).

Gordon et al. (1988) developed a quadratic algorithm between $r_{rs}(\lambda)$ and ocean's IOPs based on the reflectance model; that is,

$$r_{rs}(\lambda) = g_1 \left(\frac{b_b(\lambda)}{a(\lambda) + b_b(\lambda)} \right) + g_2 \left(\frac{b_b(\lambda)}{a(\lambda) + b_b(\lambda)} \right)^2 \quad (2)$$

Consequently, $b_b(\lambda)/(a(\lambda) + b_b(\lambda))$ can be calculated from subsurface remote-sensing reflectance

$$\left(\frac{b_b(\lambda)}{a(\lambda) + b_b(\lambda)} \right) = \frac{-g_1 + \sqrt{g_1^2 + 4g_2 r_{rs}(\lambda)}}{2g_2} \quad (3)$$

where $r_{rs}(\lambda)$ is remote-sensing reflectance just beneath the surface, g_1 and g_2 are constant values of 0.0948 and 0.0794 as derived in Gordon et al. (1988). Coefficients $a(\lambda)$ and

$b_b(\lambda)$ are total absorption and backscattering coefficients, respectively, which can be expressed as

$$a(\lambda) = a_w(\lambda) + a_{ph}(\lambda) + a_g(\lambda) + a_d(\lambda) \quad (4)$$

$$b_b(\lambda) = b_{bw}(\lambda) + b_{bp}(\lambda), \quad (5)$$

where $a_w(\lambda)$, $a_{ph}(\lambda)$, $a_g(\lambda)$, and $a_d(\lambda)$ are absorption coefficients for pure seawater, phytoplankton, dissolved matters, and non-algal particles, respectively, and $b_w(\lambda)$ and $b_{bp}(\lambda)$ are backscattering coefficients for pure seawater and particulate matters, respectively. In Eq. 5, $b_{bp}(\lambda)$ is spectrally correlated, and generally can be modeled as a function of $b_{bp}(\lambda)$ at a reference wavelength (Gordon and Morel 1983; IOCCG 2006; i.e., $b_{bp}(\lambda_0)$, with a power law slope η). The function can be expressed as

$$b_{bp}(\lambda) = b_{bp}(\lambda_0) \left(\frac{\lambda}{\lambda_0} \right)^\eta \quad (6)$$

$nL_w(\lambda)$ in the NIR wavelengths (i.e., 748, 859, and 869 nm) can be directly derived using the SWIR atmospheric correction with the SWIR 1640 and 2130 nm bands (Figs. 2, 3) in the turbid waters. In comparison to the $nL_w(\lambda)$ results in visible bands, which are usually used for IOP modeling, pure seawater absorption coefficients $a_w(\lambda)$ at the NIR bands are significantly higher than those for other seawater constituents (i.e., $a_w(\lambda) \gg a_{ph}(\lambda)$, $a_g(\lambda)$, and $a_d(\lambda)$). For example, $a_w(869)$ is $\sim 5 \text{ m}^{-1}$, compared with $a_{ph}(\lambda)$, $a_g(\lambda)$, and $a_d(\lambda)$ that are close to 0 (Babin and Stramski 2002; Stramski et al. 2007; Tassan and Ferrari

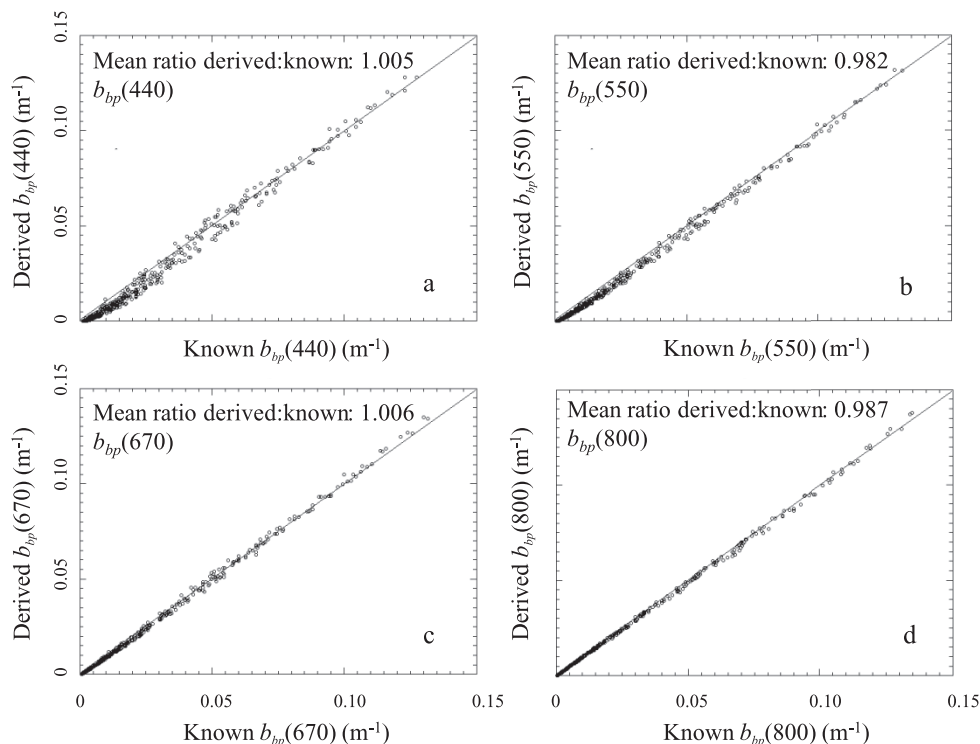


Fig. 6. Comparison between the model-derived particle backscattering coefficient $b_{bp}(\lambda)$ and the true values from the IOCCG synthetic data set for (a) $b_{bp}(440)$, (b) $b_{bp}(550)$, (c) $b_{bp}(670)$, and (d) $b_{bp}(800)$.

2003). Therefore, with the spectral features of $a_w(\lambda)$, $a_{ph}(\lambda)$, $a_g(\lambda)$, and $a_d(\lambda)$ at the NIR wavelengths, $b_b(\lambda)/(a(\lambda) + b_b(\lambda))$ can be approximated as

$$\left(\frac{b_b(\lambda)}{a(\lambda) + b_b(\lambda)} \right) \approx \left(\frac{b_b(\lambda)}{a_w(\lambda) + b_b(\lambda)} \right) \quad (7)$$

Thus, $b_{bp}(\lambda)$ for the NIR wavelengths can be derived using Eqs. 3, 5, and 7 from $r_{rs}(\lambda)$ in the NIR wavelengths. Consequently, power law slope η can be calculated with $b_{bp}(748)$ and $b_{bp}(869)$ using Eq. 6. With the derived η and a reference $b_{bp}(748)$ (or $b_{bp}(869)$), $b_{bp}(\lambda)$ in the MODIS-Aqua visible bands can be calculated using Eq. 6. Finally, $a(\lambda)$ in the MODIS-Aqua visible bands can be derived using Eq. 3 from satellite-measured $nL_w(\lambda)$ spectra data in the visible.

Test with synthetic data set: The IOP modeling using the visible $nL_w(\lambda)$ in the turbid waters has been a challenge because of the complexity of the seawater constituents as well as the dynamic variability in the coastal environment. The NIR-based IOP modeling approach as described in Eqs. 2–7 has the potential to derive accurate satellite IOP data in turbid waters, and provides much potential in further research. Actually, initial tests with the International Ocean Colour Coordinating Group (IOCCG) Synthesized Data set for IOP algorithm development from IOCCG Report 5 (IOCCG 2006) show that $b_{bp}(\lambda)$ in the visible bands can be reasonably derived using the NIR ocean reflectance as formulated above.

Figure 6 shows the $b_{bp}(\lambda)$ retrievals at the wavelengths of 440, 550, 670, and 800 nm using the NIR-based IOP

modeling approach. With $r_{rs}(800)$ and $r_{rs}(740)$ values provided in the IOCCG data set (IOCCG 2006), $b_{bp}(800)$ (Fig. 6d) and $b_{bp}(740)$ (not shown here) can be computed using Eqs. 3, 5, and 7. Consequently, power law slope η can be derived, and power law slope η and $b_{bp}(800)$ are then used to compute $b_{bp}(440)$ (Fig. 6a), $b_{bp}(550)$ (Fig. 6b), and $b_{bp}(670)$ (Fig. 6c) using Eq. 6. The derived $b_{bp}(440)$, $b_{bp}(550)$, $b_{bp}(670)$, and $b_{bp}(800)$ are generally consistent with the known (true) values for the entire data set (particularly for turbid waters) even though some biases do exist in the clear waters with $R_{rs}(740)$ close to 0.

Considering the near-zero NIR $\rho_{wN}(\lambda)$ in the open ocean and the retrieval uncertainty, as well as noise in satellite-derived NIR $\rho_{wN}(\lambda)$, the NIR-based IOP approach can only be applied practically to the turbid and highly turbid waters with significant NIR $nL_w(\lambda)$ contributions. Specifically, $b_{bp}(800)$ and $b_{bp}(440)$ in the range of ~ 0.06 – 0.14 m^{-1} are derived from the data with $R_{rs}(740) > 0.001 \text{ sr}^{-1}$ (71 points), which correspond to $K_d(490) > \sim 1.0 \text{ m}^{-1}$, and fall into the category of turbid waters as defined in Shi and Wang (2010a). The derived $b_{bp}(440)$, $b_{bp}(550)$, $b_{bp}(670)$, and $b_{bp}(800)$ for $R_{rs}(740) > 0.001 \text{ sr}^{-1}$ match quite well with true $b_{bp}(\lambda)$ data. The mean ratios of derived and true $b_{bp}(\lambda)$ for the data with $R_{rs}(740) > 0.001 \text{ sr}^{-1}$ are 1.005, 0.982, 1.006, 0.983, and 0.987 for $b_{bp}(440)$, $b_{bp}(550)$, $b_{bp}(670)$, $b_{bp}(740)$, and $b_{bp}(800)$, respectively. The corresponding standard deviation values for retrievals of $b_{bp}(440)$, $b_{bp}(550)$, $b_{bp}(670)$, $b_{bp}(740)$, and $b_{bp}(800)$ are 3.0×10^{-3} , 1.9×10^{-3} , 1.6×10^{-3} , 1.5×10^{-3} , and $1.6 \times 10^{-3} \text{ m}^{-1}$, respectively.

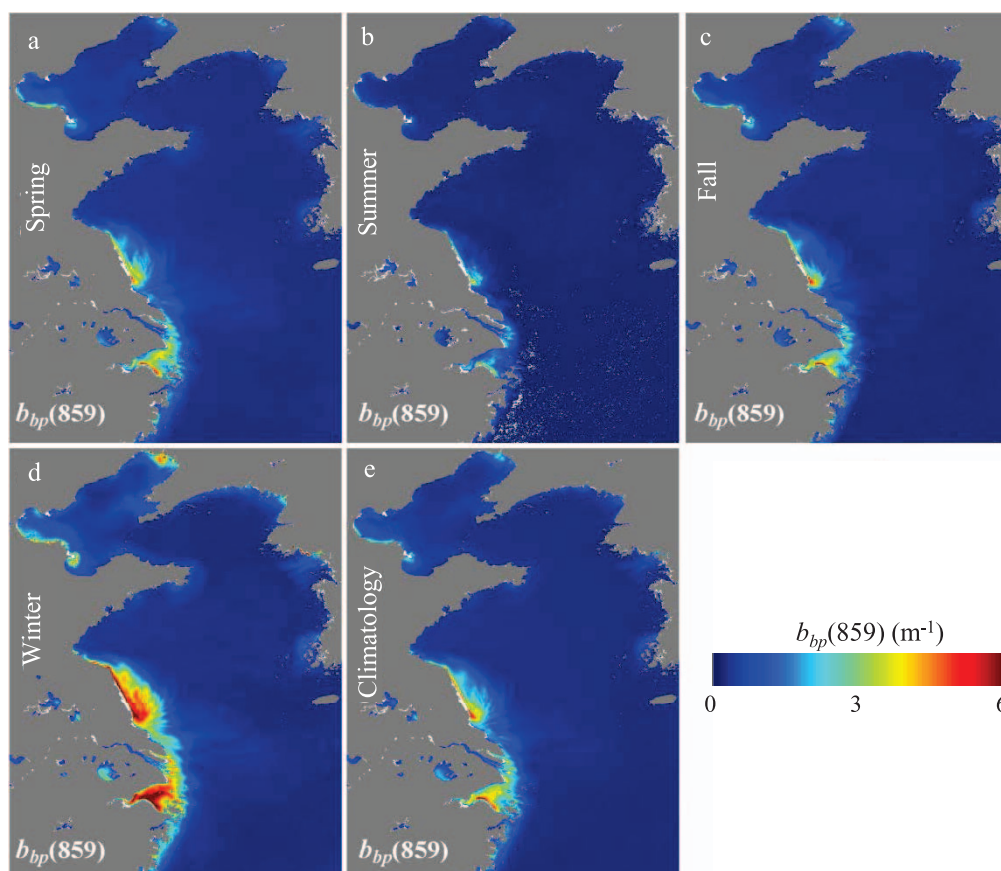


Fig. 7. Three-year mean images of backscattering coefficient at the wavelength of 859 nm $b_{bp}(859)$ derived from MODIS-Aqua measurements from 2009 to 2011 for (a) spring (March–May), (b) summer (June–August), (c) autumn (September–November), (d) winter (December–February), and (e) 3 yr climatology.

The other IOP parameters such as $a(\lambda)$ can also be derived in the turbid waters using $r_{rs}(\lambda)$ values in the visible bands following Eq. 3 with derived $b_{bp}(\lambda)$ in the visible. For satellite ocean-color remote sensing, this evidently shows that particle backscattering spectrum $b_{bp}(\lambda)$ can be reasonably derived analytically from satellite $nL_w(\lambda)$ measurements in the NIR wavelengths in the turbid and highly turbid waters, and correspondingly other IOP parameters such as $a(\lambda)$ can be computed from $nL_w(\lambda)$ measurements in the visible. Because of the inaccessibility to the in situ measurements of reflectance and IOP properties in these highly turbid ocean regions, we are not able to conduct detailed evaluation and validation with the in situ data on the NIR-based IOP algorithm in the BS, YS, and ECS. Even though coastal waters in the BS, YS, and ECS are generally more turbid than cases with the IOCCG test data set, the robust approach with the analytical formula and the accurate retrievals of $b_{bp}(\lambda)$ with the IOCCG synthetic data set suggest that IOP properties in the BS, YS, and ECS can be retrieved from the NIR-based IOP approach.

Demonstration of the NIR-based IOP retrievals in the three seas: Figure 7 shows the seasonal change and mean climatology of $b_{bp}(859)$ between 2009 and 2011 in the BS, YS, and ECS. Apparently, $b_{bp}(859)$ in the coastal regions are much larger than those in open oceans. In general, the spatial patterns of $b_{bp}(859)$ are similar to, and consistent

with, $nL_w(859)$ distributions in Fig. 2. Enhanced $b_{bp}(859)$ can be found in the Subei Shoal of the BS, Yangtze River Estuary, and Hangzhou Bay (Fig. 7e). Seasonal variation of $b_{bp}(859)$ is significant in these highly turbid regions. For example, $b_{bp}(859)$ in Hangzhou Bay is $\sim 1\text{--}2\text{ m}^{-1}$ in the summer (Fig. 7b), while it reaches over $\sim 5\text{ m}^{-1}$ during the winter season (Fig. 7d). These values of $b_{bp}(859)$ are actually in the same order as corresponding $a_w(859)$, and are considerably higher than those in the IOCCG synthetic data set as shown in Fig. 6.

With $b_{bp}(748)$ and $b_{bp}(869)$ (or $b_{bp}(859)$), power law slope η can be derived in the highly turbid waters in order to compute $b_{bp}(\lambda)$ in the visible. Figure 8 further details the variation of η corresponding to the change of $b_{bp}(859)$ at different stations in the BS, YS, and ECS. Similar to the scatter plots of $nL_w(748)$ vs. $nL_w(869)$, large uncertainties of power law slope η occur when values of $b_{bp}(859)$ are small (as expected). Thus, some large positive η values might be attributed to sensor-low SNR at the SWIR bands as well as errors from the SWIR atmospheric correction for low $b_{bp}(859)$ (or $nL_w(859)$) values, and are not valid. In fact, for low $b_{bp}(859)$ values, it is simply not sensitive (or accurate) for computing η values (i.e., errors in η are huge for low $b_{bp}(859)$ values). However, both $b_{bp}(859)$ and η values at these stations are generally consistent with the in situ observations in the other turbid waters (Reynolds et al.

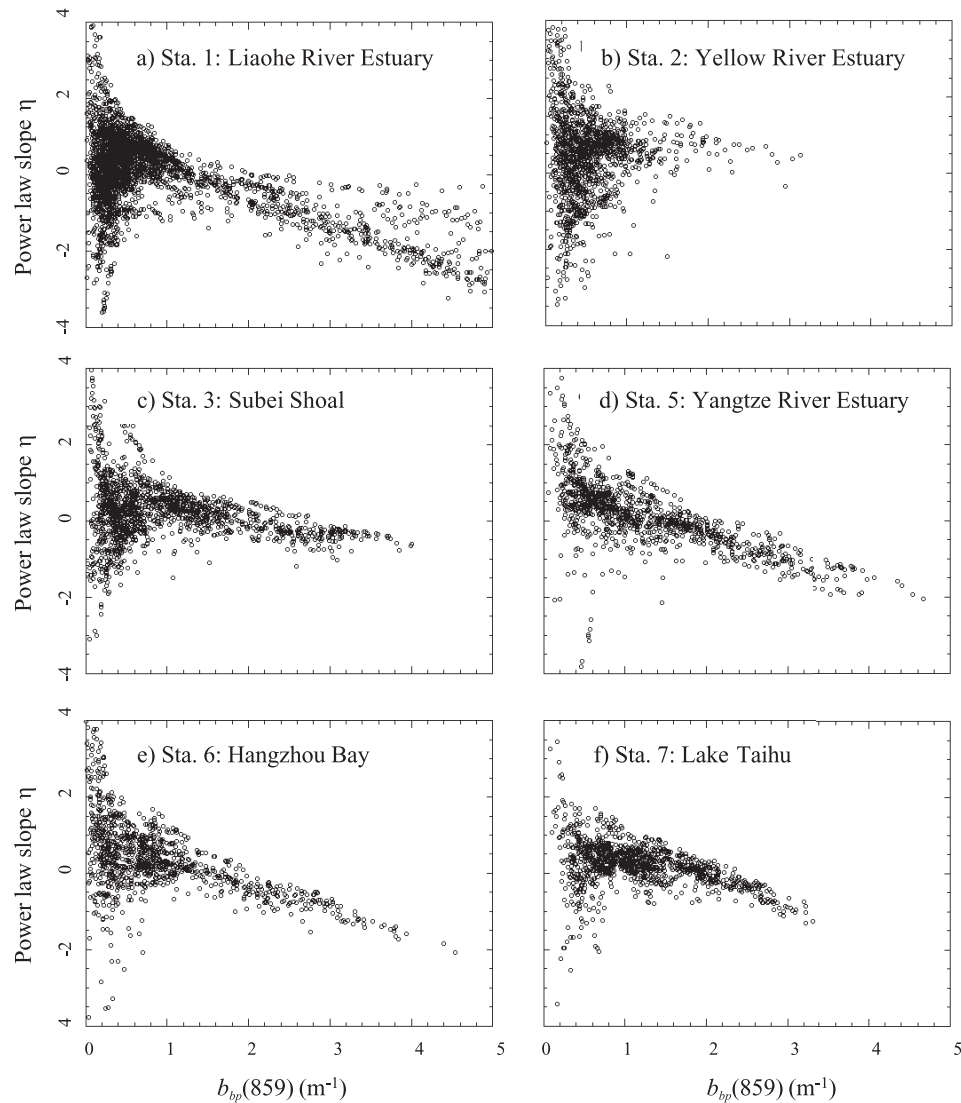


Fig. 8. Scatter plots of MODIS-Aqua-derived power law slope η vs. $b_{bp}(859)$ from 2009 to 2011 in a 5×5 box centered at (a) Sta. 1, (b) Sta. 2, (c) Sta. 3, (d) Sta. 5, (e) Sta. 6, and (f) Sta. 7.

2001; Doxaran et al. 2007). It is noted that for some extremely turbid cases, power law slope η data cannot be derived in Fig. 8 because the MODIS 748 nm (also 869 nm) band is saturated. Normally these cases occurred in Hangzhou Bay (Sta. 6) and Subei Shoal (Sta. 3) when $\rho_{wN}(859) > \sim 0.1$ and $b_{bp}(859) > \sim 4 \text{ m}^{-1}$.

For Sta. 2 in the Yellow River Estuary and Sta. 3 in the Subei Shoal region, η stays relatively stable with the change of $b_{bp}(859)$. However, for Sta. 5–7, η trends downward. This suggests that both $b_{bp}(859)$ and power law slope η are quite dynamic. η not only changes with the variation of $b_{bp}(859)$, but also shows regional differences with different sediment sources such as between Sta. 3 at the Subei Shoal and Sta. 5 in Hangzhou Bay. On the other hand, the trend of η might also have an important geophysical implication. Studies have shown that power law slope η is related to the particle size distribution (Loisel et al. 2006; Doxaran et al. 2007; Kostadinov et al. 2009). It is noted that the b_{bp} spectral slope (η) climatology image (2009–2011) for the

entire region (results not shown) indeed shows minimum values (< -2) in the offshore region, while η increases to ~ 0 in the coastal region and river estuary. This observation is consistent with other studies (Loisel et al. 2006).

At Sta. 5 and 6, the downward trend of η suggests less coarse particles in the water column as $b_{bp}(859)$ increases (i.e., higher TSM concentration). In comparison, η is stable at Sta. 3 for all $b_{bp}(859)$ ranging from 0 m^{-1} to 4 m^{-1} . Note that high $b_{bp}(859)$ with minimum η (~ -2) at Sta. 5 and 6 normally occurs in the winter season, while low $b_{bp}(859)$ values with η up to ~ 0 occur during the summer season. Indeed, mean $b_{bp}(859)$ and η values at the Yangtze River Estuary (Sta. 5) are 0.55 m^{-1} and 0.53 in the summer and 1.88 m^{-1} and -0.24 in the winter, while in Hangzhou Bay (Sta. 6) these values are 0.31 m^{-1} and 0.43 in the summer and 2.17 m^{-1} and -0.26 in the winter. In fact, the maximum values of $b_{bp}(859)$ at Sta. 5 and 6 in the winter can reach 4.67 m^{-1} and 4.54 m^{-1} , respectively, with the corresponding η values of -2.05 and -2.06 , respectively.

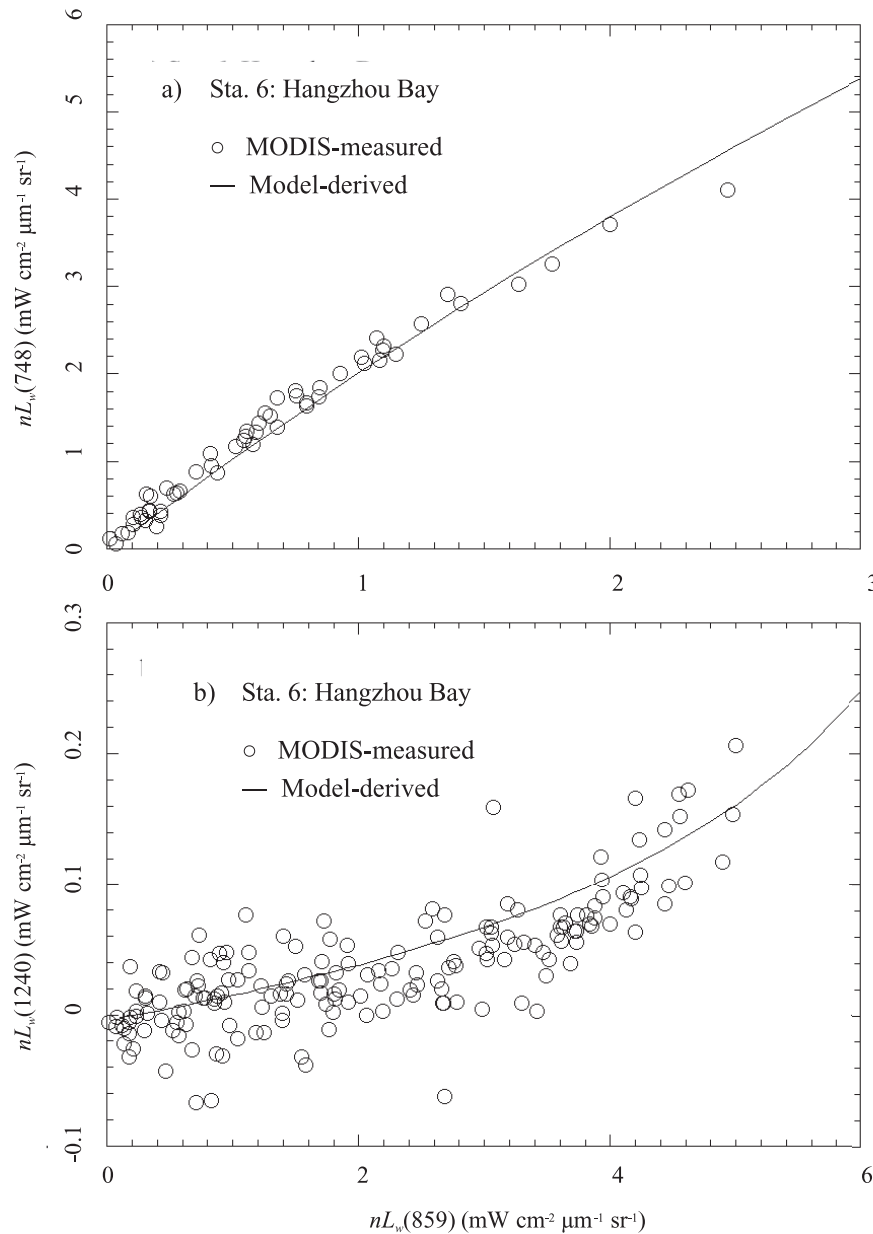


Fig. 9. Comparisons between the reflectance-model-derived and MODIS-Aqua-measured $nL_w(\lambda)$ relationships at Sta. 6 in Hangzhou Bay for (a) $nL_w(748)$ vs. $nL_w(869)$ and (b) $nL_w(859)$ vs. $nL_w(1240)$.

However, in the summer, the maximum values of $b_{bp}(859)$ at Sta. 5 and 6 are only 1.63 m^{-1} and 1.13 m^{-1} , respectively, with the corresponding η values of -0.01 and 0.12 , respectively. This indicates that seasonal change of Yangtze River discharge and river plume can also affect the particle size distribution in addition to the process of the sediment re-suspension at these two stations. One possible explanation is that Yangtze River brings in fine particles to the Yangtze River Estuary and Hangzhou Bay during the dry season in the winter, while the water column in the Yangtze River Estuary and Hangzhou Bay contains more coarse particles due to large river runoff during the wet season in the summer. Indeed, field observations in the Yangtze River Estuary show that there is a remarkable

seasonal cycle in the grain size distribution, with the smallest particle size in the winter and spring and largest particles in the summer and autumn (Yang et al. 2008). This further suggests that river discharge and river plume may play a more important role in determining the particle size distribution than the other physical processes at Sta. 5 and 6.

Figure 9 provides further evidence that $b_{bp}(748)$ and $b_{bp}(869)$, as well as the power law slope η derived from the NIR ocean-reflectance model, are consistent with satellite measurements and can be applied in further IOP modeling computation. At Sta. 6 in Hangzhou Bay, the 3 yr mean value of η as shown in Fig. 8e is ~ -0.12 . In fact, using Eqs. 1, 2, 5, and 6, $\rho_{wN}(748)$, $\rho_{wN}(1240)$, and $\rho_{wN}(1640)$ can

be analytically derived from $\rho_{wN}(859)$ (or $\rho_{wN}(869)$). The modeled relationship between $nL_w(748)$ and $nL_w(869)$ is compared with MODIS-Aqua-derived $nL_w(748)$ and $nL_w(869)$ (Fig. 9a). In general, they match well with each other for a wide range of $nL_w(869)$ from $0 \text{ mW cm}^{-2} \mu\text{m}^{-1} \text{ sr}^{-1}$ to $2.5 \text{ mW cm}^{-2} \mu\text{m}^{-1} \text{ sr}^{-1}$. This evidently shows that the relationship between $nL_w(748)$ and $nL_w(869)$ can be accurately characterized with a priori knowledge of power law slope η . Tests of the $nL_w(748)$ and $nL_w(869)$ at the other stations show similar results to those from Sta. 6.

Similar to Fig. 9a, the modeled function between $nL_w(859)$ and $nL_w(1240)$ matches reasonably well with satellite-derived $nL_w(859)$ and $nL_w(1240)$ data at Sta. 6 in Hangzhou Bay (Fig. 9b). The discrepancy between modeled $nL_w(\lambda)$ and satellite retrievals in Fig. 9 can be attributed to the SWIR sensor noise and uncertainties associated with atmospheric correction, as well as uncertainties from IOP and reflectance models. On the other hand, modeled $nL_w(1640)$ (not shown here) as a function of $nL_w(859)$ shows that $nL_w(1640)$ is essentially close to 0 even for extremely turbid waters with $nL_w(859)$ over $\sim 5 \text{ mW cm}^{-2} \mu\text{m}^{-1} \text{ sr}^{-1}$. In comparison, the MODIS SWIR 1240 nm band is only black for waters with $nL_w(859)$ less than $\sim 2.5 \text{ mW cm}^{-2} \mu\text{m}^{-1} \text{ sr}^{-1}$. This further shows that MODIS SWIR 1640 and 2130 nm bands are black for all types of turbid waters, confirming that the SWIR band set of 1640 and 2130 nm can be confidently used to derive satellite ocean-color products in other highly turbid regions such as Amazon River Estuary, La Plata River Estuary, and Lake Taihu (Shi and Wang 2009a; Wang et al. 2013b). The consistency between satellite-measured and the model-derived $nL_w(\lambda)$ values in Fig. 9 provides further evidence that values of power law slope η derived from the NIR ocean reflectance in turbid waters are reliable and can be consequently used to derive the other IOP products.

Most currently existing IOP models are empirical or semi-analytical. Even though power law of $b_{bp}(\lambda)$ (Eq. 6) has been widely used in the IOP modeling efforts, it typically has been a challenge to accurately estimate reference $b_{bp}(\lambda_0)$ and η directly with the ocean reflectance in the visible bands. In the Garver–Siegel–Maritorena IOP model (Garver and Siegel 1997; Maritorena et al. 2002), η is set to a constant value. On the other hand, $b_{bp}(\lambda_0)$ and η in Quasi-Analytical Algorithm (Lee et al. 2002) are empirically computed using ocean reflectance in the visible bands. Sensitivity studies have found that η can significantly affect the accuracy of IOP retrievals (Hoge and Lyon 1996). In comparison, retrievals of ocean IOP data in this study are basically analytical because they are based on the spectral features of the seawater constituents in the NIR wavelengths. This analytical feature of IOP modeling with the NIR ocean reflectance indicates that it can address the large variation of η values (Reynolds et al. 2001; Wozniak and Stramski 2004), which otherwise are expected to lead to large errors with empirical or semi-analytical approaches.

Implications for ocean geo-parameter retrievals—In coastal regions, several satellite blue-green ratio algorithms for biological and biogeochemical products such as Chl *a* and $K_d(490)$ can have significant biases. Consequently, regional algorithms using $nL_w(\lambda)$ in the red band are proposed for

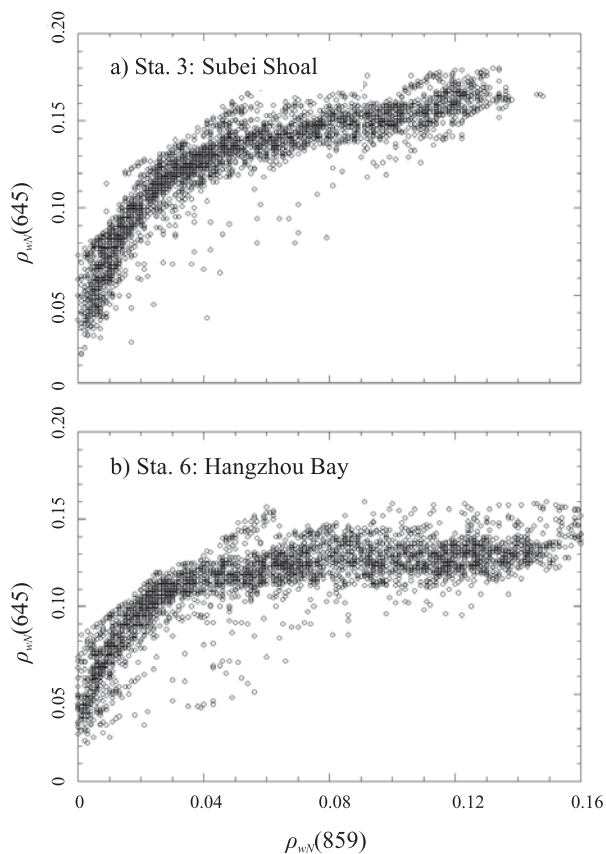


Fig. 10. Scatter plots of MODIS-Aqua-derived $\rho_{wN}(645)$ vs. $\rho_{wN}(859)$ from 2009 to 2011 in a 5×5 box centered at (a) Sta. 3 in the Subei Shoal and (b) Sta. 6 in Hangzhou Bay.

estimating Chl *a* (Dall’olmo et al. 2005), TSM concentration (Zhang et al. 2010; Son and Wang 2012), and $K_d(490)$ [Lee et al. 2005; Doron et al. 2007; Wang et al. 2009a]. As suggested in Eq. 2, there are limitations of $\rho_{wN}(\lambda)$ at both red and NIR wavelengths. Figure 10 further shows the maximal limitation of $\rho_{wN}(645)$ at Sta. 3 in the Subei Shoal (Fig. 10a) and at Sta. 6 in Hangzhou Bay (Fig. 10b). At these two stations, $\rho_{wN}(645)$ generally co-varies with $\rho_{wN}(859)$ when $\rho_{wN}(859)$ is less than ~ 0.05 ($nL_w(859) \sim 1.5 \text{ mW cm}^{-2} \mu\text{m}^{-1} \text{ sr}^{-1}$), while $\rho_{wN}(645)$ is less sensitive (or insensitive) to the change of $\rho_{wN}(859)$ for $\rho_{wN}(859)$ over ~ 0.05 . Essentially, $\rho_{wN}(645)$ becomes more or less a constant at Sta. 6 in Hangzhou Bay for $\rho_{wN}(859)$ over ~ 0.05 . In addition, curves for $\rho_{wN}(645)$ vs. $\rho_{wN}(859)$ at Sta. 3 and 6 are not exactly the same, i.e., for the same $\rho_{wN}(859)$, $\rho_{wN}(645)$ values at Sta. 3 and 6 are different. This indicates that IOP properties at these two stations are also different. Other constituents, such as Chl *a*, might also play significant roles in the red (645 nm) even though their effects are negligible in the NIR wavelengths.

Figure 10 provides evidence that $\rho_{wN}(645)$ becomes less related or even totally unrelated to the sea water constituent changes when $\rho_{wN}(859)$ is over ~ 0.05 , suggesting that satellite ocean-color algorithms (e.g., Chl *a* algorithms [Dall’olmo et al. 2005; Gitelson et al. 2007]), TSM algorithms (Miller and McKee 2004; Zhang et al. 2010;

Son and Wang 2012), $K_d(490)$ algorithms (Lee et al. 2005; Wang et al. 2009a), and IOP algorithms (Lee et al. 2002) using $nL_w(\lambda)$ at the red band have some significant limitations in deriving water properties in highly turbid waters with $\rho_{wN}(859)$ over ~ 0.05 . The reflectance saturation phenomenon is attributed to the dominance of $b_b(\lambda)$ for extremely turbid waters in a function of $b_b(\lambda)/(a(\lambda) + b_b(\lambda))$, of which $\rho_{wN}(\lambda)$ can be expressed as a function (Gordon et al. 1988). When $b_b(\lambda) \gg a(\lambda)$, $\rho_{wN}(\lambda)$ reaches its maximum value (Shi and Wang 2009b; Wang et al. 2013b). Accurate retrievals are expected only when $\rho_{wN}(859) < \sim 0.05$. For turbid waters with $\rho_{wN}(859)$ over ~ 0.05 , ocean reflectance $\rho_{wN}(\lambda)$ in the NIR wavelengths is required in order to derive accurate satellite ocean-color products. This further demonstrates that the NIR ocean reflectance in the coastal region is a prerequisite in order to derive ocean biological and biogeochemical parameters in highly turbid waters.

This assessment for the NIR ocean reflectance in highly turbid waters is consistent with the conclusions from in situ optics measurements (Shen et al. 2010; Zhang et al. 2012). In the Yangtze River Estuary, Shen et al. (2010) proposed to use multiple $R_{rs}(\lambda)$ from the green to NIR with the corresponding $R_{rs}(\lambda)$ to retrieve suspended sediment concentration levels in the Yangtze River Estuary and Hangzhou Bay to avoid situations of $R_{rs}(\lambda)$ saturation (or reduced $R_{rs}(\lambda)$ sensitivity) at certain wavelengths. In the highly turbid Lake Taihu, Zhang et al. (2012) found poor estimations of $K_d(\text{PAR})$ derived from all existing empirical and semi-analytical models using the reflectance in the visible wavelengths. Instead, a simple optical model from the NIR $R_{rs}(\lambda)$ at wavelengths of 748 and 859 nm shows good accuracy for $K_d(\text{PAR})$ estimation.

In addition, Fig. 10 also has a specific implication for atmospheric correction for satellite ocean-color remote sensing. Iterative approaches with ocean reflectance in the red and NIR bands for estimating the NIR ocean contribution have long been used (Stumpf et al. 2003; Bailey et al. 2010; Wang et al. 2013a). However, results in Fig. 10 show that $\rho_{wN}(645)$ is insensitive to the change of the NIR ocean reflectance when $\rho_{wN}(859)$ is over ~ 0.05 ($nL_w(859) \sim 1.5 \text{ mW cm}^{-2} \mu\text{m}^{-1} \text{ sr}^{-1}$). Thus, the NIR ocean reflectance can only be accurately estimated using the iterative approach for turbid waters with $\rho_{wN}(859) < \sim 0.05$. Because of this, atmospheric correction with the satellite NIR measurements and the modification for removing NIR ocean contributions using ocean reflectance in the red band is not valid in highly turbid waters. This provides further evidence that the SWIR atmospheric correction algorithm for the satellite ocean-color data processing is indispensable in order to derive accurate $nL_w(\lambda)$ spectra data for highly turbid waters such as the coastal waters in the BS, YS, and ECS, as well as in Lake Taihu.

This study quantifies and characterizes the normalized water-leaving radiance spectra $nL_w(\lambda)$ in wavelengths of the red, NIR, and SWIR in coastal regions of the BS, YS, and ECS. Enhanced $nL_w(645)$ and $nL_w(859)$ are found in the Subei Shoal of the YS and Hangzhou Bay of the ECS. Three-year mean $nL_w(859)$ can reach over $\sim 3 \text{ mW cm}^{-2} \mu\text{m}^{-1} \text{ sr}^{-1}$ for a large portion of Hangzhou Bay. Significant seasonal changes of $nL_w(\lambda)$ in the red,

NIR, and SWIR wavelengths can be found in the coastal region of the BS, YS, and ECS. Winter and summer are the two seasons with the most and least enhanced red and NIR $nL_w(\lambda)$, respectively. Even though for highly turbid waters, the black ocean assumption is still valid for MODIS SWIR 1640 and 2130 nm bands, $nL_w(\lambda)$ at the SWIR 1240 nm band ($nL_w(1240)$) is not always negligible for Hangzhou Bay and Subei Shoal regions. This is particularly true for the winter season, when $nL_w(1240)$ can reach up to $\sim 0.2 \text{ mW cm}^{-2} \mu\text{m}^{-1} \text{ sr}^{-1}$ in Hangzhou Bay.

This study also shows the highly dynamic feature of the NIR ocean-reflectance spectral shape represented by reflectance ratio of $\rho_{wN}(748) : \rho_{wN}(869)$ in a large coverage of turbid waters such as the western Pacific region. The quasi-linear relationship between $\rho_{wN}(748)$ and $\rho_{wN}(869)$ in the less turbid waters is intrinsically attributed to pure water absorption ($a_w(\lambda)$) and particle backscattering ($b_{bp}(\lambda)$) at wavelengths of 748 and 869 nm (Ruddick et al. 2006). Variations in $\rho_{wN}(748) : \rho_{wN}(869)$ are shown to be regional-dependent. We provide the evidence as results in Figs. 3 and 4, showing that satellite-observed NIR spectral-reflectance features such as the $\rho_{wN}(748)$ vs. $\rho_{wN}(869)$ relationship are regionally different and might be related to the sediment properties (e.g., particle size distribution, compositions, and textures). The NIR spectral-reflectance features with the same sediment source are similar in the three seas. However, the NIR reflectance features with different sediment sources (e.g., between those from the Yellow River and Ancient Yellow River at Sta. 2 and 3 and from Yangtze River at Sta. 5–7) are noticeably different. For certain turbid regions, reflectance ratio $\rho_{wN}(748) : \rho_{wN}(869)$ decreases notably with the increase of the NIR ocean reflectance. It has been shown that there is no universal constant of $\rho_{wN}(748) : \rho_{wN}(869)$ due to the large dynamic range of the NIR $\rho_{wN}(\lambda)$, complexity of seawater constituents and their optical properties, as well as the significant seasonal variability in highly turbid waters of the BS, YS, and ECS. This assessment is consistent with the results from Doron et al. (2011), which show that the shape of ocean-reflectance spectrum in the NIR varies in the turbid to highly turbid waters. The flattening of the NIR reflectance spectrum with increasing sediment concentration in the water column can be mainly attributed to the gradual increase in the contribution of suspended sediments to $nL_w(\lambda)$ in the NIR wavelengths (Doron et al. 2011).

We demonstrate that IOP properties such as $b_{bp}(\lambda)$ can be derived from satellite $nL_w(\lambda)$ measurements in the NIR wavelengths. Backscattering spectral slopes can also be reasonably derived in the coastal turbid regions. In the BS, YS, and ECS, the NIR $b_{bp}(\lambda)$ shows significant seasonal variability. $b_{bp}(\lambda)$ in the winter is much higher than that in the other seasons. Backscattering spectral slopes for different sediment sources also have different features. We show that the power law slope η is quite dynamic and can have different features for different regions (i.e., different particle sources).

It should be noted that the MODIS two-NIR bands at 748 and 869 nm can be saturated for some extremely turbid stations such as Sta. 6 in Hangzhou Bay. For example, the maximal derived $\rho_{wN}(869)$ is ~ 0.09 , while $\rho_{wN}(859)$; band 859 nm does not saturate) can reach ~ 0.16 . Thus, the NIR

spectral reflectance results in Figs. 3f, 4d, and 8e only represent part of the scenarios in Hangzhou Bay.

This study also characterizes the $nL_w(1240)$ variations in highly turbid waters, and shows that the SWIR atmospheric correction with the MODIS SWIR band set of 1240 and 2130 nm can be safely used for $nL_w(\lambda)$ retrievals for waters with $nL_w(859)$ less than $\sim 2.5 \text{ mW cm}^{-2} \mu\text{m}^{-1} \text{ sr}^{-1}$. On the other hand, there are non-negligible $nL_w(1240)$ contributions for waters with $nL_w(859) > \sim 2.5 \text{ mW cm}^{-2} \mu\text{m}^{-1} \text{ sr}^{-1}$, which can lead to biased $nL_w(\lambda)$ retrievals. This suggests that in highly turbid waters such as Hangzhou Bay and Amazon River (Shi and Wang 2010a), atmospheric correction using the SWIR band set of 1640 and 2130 nm is necessary in order to derive accurate $nL_w(\lambda)$ spectra in the visible, NIR, and SWIR 1240 nm.

Specific features of the MODIS-Aqua-measured $\rho_{wN}(645)$ in this study show that $\rho_{wN}(645)$ becomes less sensitive or even insensitive to the change of ocean environments for $\rho_{wN}(859) > \sim 0.05$, implying that the current existing algorithms for Chl *a*, $K_d(490)$, TSM, and IOPs using the ocean reflectance at the red band for coastal regions are all limited and cannot be applied to highly turbid waters with $\rho_{wN}(859) > \sim 0.05$. In addition, the optical feature of insensitive ocean reflectance in red band $\rho_{wN}(645)$ implies that the current National Aeronautics and Space Administration standard satellite ocean-color atmospheric correction algorithm, which uses red ocean reflectance to estimate the NIR ocean reflectance (Bailey et al. 2010), may not be applicable for highly turbid waters. The SWIR atmospheric correction algorithm is required to derive accurate satellite water property products in ocean regions such as the Subei Shoal, Yangtze River Estuary, La Plata River Estuary, and Hangzhou Bay.

Based on the fact that the pure seawater absorption in the NIR wavelengths is much larger than those of other constituents in turbid waters, ocean IOP modeling can be significantly simplified. This study demonstrated that it is feasible to analytically derive some IOP properties in turbid waters with combined visible-NIR $nL_w(\lambda)$ spectra data. This NIR-based IOP modeling approach is different from the currently existing IOP models. Although further investigation is still needed, results from this study indeed suggest that the NIR-based IOP modeling approach is promising and has a great potential for accurately characterizing and quantifying IOP properties in turbid waters such as in the BS, YS, and ECS.

Acknowledgments

The Moderate Resolution Imaging Spectroradiometer (MODIS) Level-1B data were obtained from the National Aeronautics and Space Administration Goddard Space Flight Center MODIS Adaptive Processing System Services website. We thank two anonymous reviewers for their useful comments. The views, opinions, and findings contained in this paper are those of the authors and should not be construed as an official National Oceanic and Atmospheric Administration or U.S. Government position, policy, or decision.

References

BABIN, M., A. MOREL, V. FOURNIER-SICRE, F. FELL, AND D. STRAMSKI. 2003. Light scattering properties of marine particles in coastal and open ocean waters as related to the particle mass concentration. *Limnol. Oceanogr.* **48**: 843–859, doi:10.4319/lo.2003.48.2.0843

———, AND D. STRAMSKI. 2002. Light absorption by aquatic particles in the near-infrared spectral region. *Limnol. Oceanogr.* **47**: 911–915, doi:10.4319/lo.2002.47.3.0911

BAI, Y., D. PAN, W.-J. CAI, X. HE, D. WANG, B. TAO, AND Q. ZHU. 2013. Remote sensing of salinity from satellite-derived CDOM in the Changjiang River-dominated East China Sea. *J. Geophys. Res.* **118**: 227–243, doi:10.1029/2012JC008467

BAILEY, S. W., B. A. FRANZ, AND P. J. WERDELL. 2010. Estimation of near-infrared water-leaving reflectance for satellite ocean color data processing. *Opt. Express* **18**: 7521–7527, doi:10.1364/OE.18.007521

DAL'OLMO, G., A. A. GITELSON, D. C. RUNDQUIST, B. LEAVITT, T. BARROW, AND J. C. HOLZ. 2005. Assessing the potential of SeaWiFS and MODIS for estimating chlorophyll concentration in turbid productive waters using red and near-infrared bands. *Remote Sens. Environ.* **96**: 176–187, doi:10.1016/j.rse.2005.02.007

DORON, M., M. BABIN, A. MANGIN, AND O. HEMBISE. 2007. Estimation of light penetration, and horizontal and vertical visibility in oceanic and coastal waters from surface reflectance. *J. Geophys. Res.* **112**: C06003, doi:10.1029/2006JC004007

———, S. BELANGER, D. DOXARAN, AND M. BABIN. 2011. Spectral variations in the near-infrared ocean reflectance. *Remote Sens. Environ.* **115**: 1617–1631, doi:10.1016/j.rse.2011.01.015

DOXARAN, D., M. BABIN, AND E. LEYMARIE. 2007. Near-infrared light scattering by particles in coastal waters. *Opt. Express* **15**: 12834–12849, doi:10.1364/OE.15.012834

———, J. M. FROIDEFOND, S. LAVENDER, AND P. CASTAING. 2002. Spectral signature of highly turbid waters—Application with SPOT data to quantify suspended particulate matter concentrations. *Remote Sens. Environ.* **81**: 149–161, doi:10.1016/S0034-4257(01)00341-8

FANG, G. H. 1994. Tides and tidal currents in East China Sea, Huanghai Sea and Bohai Sea, p. 101–112. *In* D. Zhou, Y. B. Liang, and C. K. Zeng [eds.] [ed], *Oceanology of China Seas*. Kluwer Academic.

GARVER, S. A., AND D. A. SIEGEL. 1997. Inherent optical property inversion of ocean color spectra and its biogeochemical interpretation. 1. Time series from the Sargasso Sea. *J. Geophys. Res.* **102**: 18607–18625, doi:10.1029/96JC03243

GITELSON, A. A., J. F. SCHALLES, AND C. M. HLADIK. 2007. Remote chlorophyll-*a* retrieval in turbid, productive estuaries: Chesapeake Bay case study. *Remote Sens. Environ.* **109**: 464–472, doi:10.1016/j.rse.2007.01.016

GONG, G.-C. 2004. Absorption coefficients of colored dissolved organic matter in the surface waters of the East China Sea. *Terr. Atmos. Oceanic Sci.* **15**: 75–87.

GORDON, H. R. 2005. Normalized water-leaving radiance: Revisiting the influence of surface roughness. *Appl. Opt.* **44**: 241–248, doi:10.1364/AO.44.000241

———, O. B. BROWN, R. H. EVANS, J. W. BROWN, R. C. SMITH, K. S. BAKER, AND D. K. CLARK. 1988. A semianalytic radiance model of ocean color. *J. Geophys. Res.* **93**: 10909–10924, doi:10.1029/JD093iD09p10909

———, D. K. CLARK, J. L. MUELLER, AND W. A. HOVIS. 1980. Phytoplankton pigments from the Nimbus-7 Coastal Zone Color Scanner: Comparisons with surface measurements. *Science* **210**: 63–66, doi:10.1126/science.210.4465.63

———, AND A. MOREL. 1983. Remote assessment of ocean color for interpretation of satellite visible imagery: A review. Springer-Verlag.

———, AND M. WANG. 1994. Retrieval of water-leaving radiance and aerosol optical thickness over the oceans with SeaWiFS: A preliminary algorithm. *Appl. Opt.* **33**: 443–452, doi:10.1364/AO.33.000443

- HALE, G. M., AND M. R. QUERRY. 1973. Optical constants of water in the 200 nm to 200 μ m wavelength region. *Appl. Opt.* **12**: 555–563, doi:10.1364/AO.12.000555
- HOGUE, F. E., AND P. E. LYON. 1996. Satellite retrieval of inherent optical properties by linear matrix inversion of oceanic radiance models: An analysis of model and radiance measurement errors. *J. Geophys. Res.* **101**: 16631–16648, doi:10.1029/96JC01414
- HOVIS, W. A., AND OTHERS. 1980. Nimbus 7 Coastal Zone Color Scanner: System description and initial imagery. *Science* **210**: 60–63, doi:10.1126/science.210.4465.60
- IOCCG. 2006. Remote sensing of inherent optical properties: Fundamentals, tests of algorithms, and applications. Z. Lee [ed], Reports of the International Ocean-Colour Coordinating Group, No. 5. IOCCG, Dartmouth, Canada.
- . 2010. Atmospheric correction for remotely-sensed ocean-colour products. M. Wang [ed], Reports of the International Ocean-Colour Coordinating Group, No. 10. IOCCG, Dartmouth, Canada.
- KOSTADINOV, T. S., D. A. SIEGEL, AND S. MARITORENA. 2009. Retrieval of the particle size distribution from satellite ocean color observations. *J. Geophys. Res.* **114**: C09015, doi:10.1029/2009JC005303
- LEE, Z. P., K. L. CARDER, AND R. A. ARNONE. 2002. Deriving inherent optical properties from water color: A multiband quasi-analytical algorithm for optically deep waters. *Appl. Opt.* **41**: 5755–5772, doi:10.1364/AO.41.005755
- , C. D. MOBLEY, R. G. STEWARD, AND J. S. PATCH. 1999. Hyperspectral remote sensing for shallow waters: 2. Deriving bottom depths and water properties by optimization. *Appl. Opt.* **38**: 3831–3843, doi:10.1364/AO.38.003831
- , K. DU, AND R. ARNONE. 2005. A model for the diffuse attenuation coefficient of downwelling irradiance. *J. Geophys. Res.* **110**: C02016, doi:10.1029/2004JC002275
- LOISEL, H., J. M. NICOLAS, A. SCIANDRA, D. STRAMSKI, AND A. POTEAU. 2006. Spectral dependency of optical backscattering by marine particles from satellite remote sensing of the global ocean. *J. Geophys. Res.* **111**: C09024, doi:10.1029/2005jc003367
- LU, Y., G. ZHENG, Q. TIAN, C. LYU, AND S. SUN. 2013. Analyzing the effects of particle size on remotely sensed spectra: a study on optical properties and spectral similarity scale of suspended particulate matters in water. *Appl. Opt.* **52**: 879–888, doi:10.1364/AO.52.000879
- MARITORENA, S., D. A. SIEGEL, AND A. R. PETERSON. 2002. Optimization of a semianalytical ocean color model for global-scale applications. *Appl. Opt.* **41**: 2705–2714, doi:10.1364/AO.41.002705
- MCCLAINE, C. R. 2009. A decade of satellite ocean color observations. *Annu. Rev. Mar. Sci.* **1**: 19–42, doi:10.1146/annurev.marine.010908.163650
- MILLER, R. L., AND B. MCKEE. 2004. Using MODIS Terra 250 m imagery to map concentrations of total suspended matter in coastal waters. *Remote Sens. Environ.* **93**: 259–266, doi:10.1016/j.rse.2004.07.012
- MILLIMAN, J. D., AND R. H. MEADE. 1983. World-wide delivery of river sediment to the oceans. *J. Geology* **91**: 1–21, doi:10.1086/628741
- , R. C. BEARDSLEY, Z. S. YANG, AND K. LIMEBURNER. 1985a. Modern Huanghe derived mud on the outer shelf of the East China Sea: Identification and potential mud-transport mechanisms. *Cont. Shelf Res.* **4**: 175–188, doi:10.1016/0278-4343(85)90028-7
- , H. T. SHEN, Z. S. YANG, AND R. H. MEADE. 1985b. Transport and deposition of river sediment in the Changjiang Estuary and adjacent continental-shelf. *Cont. Shelf Res.* **4**: 37–45, doi:10.1016/0278-4343(85)90020-2
- MOREL, A., AND B. GENTILI. 1996. Diffuse reflectance of oceanic waters. 3. Implication of bidirectionality for the remote-sensing problem. *Appl. Opt.* **35**: 4850–4862, doi:10.1364/AO.35.004850
- REYNOLDS, R. A., D. STRAMSKI, AND B. G. MITCHELL. 2001. A chlorophyll-dependent semianalytical reflectance model derived from field measurements of absorption and backscattering coefficients within the Southern Ocean. *J. Geophys. Res.* **106**: 7125–7138, doi:10.1029/1999JC000311
- RUDDICK, K. G., V. D. CAUWER, Y. J. PARK, AND G. MOORE. 2006. Seaborne measurements of near infrared water-leaving reflectance: The similarity spectrum for turbid waters. *Limnol. Oceanogr.* **51**: 1167–1179, doi:10.4319/lo.2006.51.2.1167
- , F. OVIDIO, AND M. RIJKEBOER. 2000. Atmospheric correction of SeaWiFS imagery for turbid coastal and inland waters. *Appl. Opt.* **39**: 897–912, doi:10.1364/AO.39.000897
- SHEN, F., W. VERHOEF, Y. X. ZHOU, M. S. SALAMA, AND X. L. LIU. 2010. Satellite estimates of wide-range suspended sediment concentrations in Changjiang (Yangtze) estuary using MERIS data. *Estuar. Coasts* **33**: 1420–1429, doi:10.1007/s12237-010-9313-2
- SHI, W., AND M. WANG. 2009a. An assessment of the black ocean pixel assumption for MODIS SWIR bands. *Remote Sens. Environ.* **113**: 1587–1597, doi:10.1016/j.rse.2009.03.011
- , AND ———. 2009b. Green macroalgae blooms in the Yellow Sea during the spring and summer of 2008. *J. Geophys. Res.* **114**: C12010, doi:10.1029/2009JC005513
- , AND ———. 2009c. Satellite observations of flood-driven Mississippi River plume in the spring of 2008. *Geophys. Res. Lett.* **36**: L07607, doi:10.1029/2009GL037210
- , AND ———. 2010a. Characterization of global ocean turbidity from Moderate Resolution Imaging Spectroradiometer ocean color observations. *J. Geophys. Res.* **115**: C11022, doi:10.1029/2010JC006160
- , AND ———. 2010b. Satellite observations of the seasonal sediment plume in central East China Sea. *J. Mar. Syst.* **82**: 280–285, doi:10.1016/j.jmarsys.2010.06.002
- , AND ———. 2012a. Satellite views of the Bohai Sea, Yellow Sea, and East China Sea. *Prog. Oceanogr.* **104**: 35–45.
- , AND ———. 2012b. Sea ice properties in the Bohai Sea measured by MODIS-Aqua: 1. Satellite algorithm development. *J. Mar. Syst.* **95**: 32–40, doi:10.1016/j.jmarsys.2012.01.012
- , AND ———. 2012c. Sea ice properties in the Bohai Sea measured by MODIS-Aqua: 2. Study of sea ice seasonal and interannual variability. *J. Mar. Syst.* **95**: 41–49, doi:10.1016/j.jmarsys.2012.01.010
- , ———, AND L. JIANG. 2011. Spring-neap tidal effects on satellite ocean color observations in the Bohai Sea, Yellow Sea, and East China Sea. *J. Geophys. Res.* **116**: C12032, doi:10.1029/2011JC007234
- SIEGEL, D. A., M. WANG, S. MARITORENA, AND W. ROBINSON. 2000. Atmospheric correction of satellite ocean color imagery: The black pixel assumption. *Appl. Opt.* **39**: 3582–3591, doi:10.1364/AO.39.003582
- SON, S., AND M. WANG. 2012. Water properties in Chesapeake Bay from MODIS-Aqua measurements. *Remote Sens. Environ.* **123**: 163–174, doi:10.1016/j.rse.2012.03.009
- , ———, AND J. SHON. 2011. Satellite observations of optical and biological properties in the Korean dump site of the Yellow Sea. *Remote Sens. Environ.* **115**: 562–572, doi:10.1016/j.rse.2010.10.002
- STRAMSKI, D., M. BABIN, AND S. B. WOZNIAK. 2007. Variations in the optical properties of terrigenous mineral-rich particulate matter suspended in seawater. *Limnol. Oceanogr.* **52**: 2418–2433, doi:10.4319/lo.2007.52.6.2418

- STUMPF, R. P., R. A. ARNONE, R. W. GOULD, P. M. MARTINOLICH, AND V. RANSIBRAHMANAKUL. 2003. A partially coupled ocean-atmosphere model for retrieval of water-leaving radiance from SeaWiFS in coastal waters, p. 51–59. *In* S. B. Hooker and E. R. Firestone [eds.] [ed], SeaWiFS Postlaunch Tech. Rep. Ser., vol. 22, NASA Tech. Memo. 2003-206892. National Aeronautics and Space Administration Goddard Space Flight Center.
- TASSAN, S., AND G. M. FERRARI. 2003. Variability of light absorption by aquatic particles in the near-infrared spectral region. *Appl. Opt.* **42**: 4802–4810, doi:10.1364/AO.42.004802
- WANG, M. 2006. Effects of ocean surface reflectance variation with solar elevation on normalized water-leaving radiance. *Appl. Opt.* **45**: 4122–4128, doi:10.1364/AO.45.004122
- . 2007. Remote sensing of the ocean contributions from ultraviolet to near-infrared using the shortwave infrared bands: Simulations. *Appl. Opt.* **46**: 1535–1547, doi:10.1364/AO.46.001535
- , J. H. AHN, L. JIANG, W. SHI, S. SON, Y. J. PARK, AND J. H. RYU. 2013a. Ocean color products from the Korean Geostationary Ocean Color Imager (GOCI). *Opt. Express* **21**: 3835–3849, doi:10.1364/OE.21.003835
- , C. J. NIM, S. SON, AND W. SHI. 2012a. Characterization of turbidity in Florida's Lake Okeechobee and Caloosahatchee and St. Lucie estuaries using MODIS-Aqua measurements. *Water Res.* **46**: 5410–5422, doi:10.1016/j.watres.2012.07.024
- , AND W. SHI. 2005. Estimation of ocean contribution at the MODIS near-infrared wavelengths along the east coast of the U.S.: Two case studies. *Geophys. Res. Lett.* **32**: L13606, doi:10.1029/2005GL022917
- , AND ———. 2007. The NIR–SWIR combined atmospheric correction approach for MODIS ocean color data processing. *Opt. Express* **15**: 15722–15733, doi:10.1364/OE.15.015722
- , AND ———. 2012. Sensor noise effects of the SWIR bands on MODIS-derived ocean color products. *IEEE Trans. Geosci. Remote Sens.* **50**: 3280–3292, doi:10.1109/TGRS.2012.2183376
- , ———, AND L. JIANG. 2012b. Atmospheric correction using near-infrared bands for satellite ocean color data processing in the turbid western Pacific region. *Opt. Express* **20**: 741–753, doi:10.1364/OE.20.000741
- , ———, AND J. TANG. 2011. Water property monitoring and assessment for China's inland Lake Taihu from MODIS-Aqua measurements. *Remote Sens. Environ.* **115**: 841–854, doi:10.1016/j.rse.2010.11.012
- , S. SON, AND J. L. W. HARDING. 2009a. Retrieval of diffuse attenuation coefficient in the Chesapeake Bay and turbid ocean regions for satellite ocean color applications. *J. Geophys. Res.* **114**: C10011, doi:10.1029/2009JC005286
- , ———, AND W. SHI. 2009b. Evaluation of MODIS SWIR and NIR–SWIR atmospheric correction algorithm using SeaBASS data. *Remote Sens. Environ.* **113**: 635–644, doi:10.1016/j.rse.2008.11.005
- , ———, Y. ZHANG, AND W. SHI. 2013b. Remote sensing of water optical property for China's inland Lake Taihu using the SWIR atmospheric correction with 1640 and 2130 nm bands. *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.* **6**: 2505–2516, doi:10.1109/jstars.2013.2243820
- , J. TANG, AND W. SHI. 2007. MODIS-derived ocean color products along the China east coastal region. *Geophys. Res. Lett.* **34**: L06611, doi:10.1029/2006GL028599
- WOZNIAK, S. B., AND D. STRAMSKI. 2004. Modeling the optical properties of mineral particles suspended in seawater and their influence on ocean reflectance and chlorophyll estimation from remote sensing algorithms. *Appl. Opt.* **43**: 3489–3503, doi:10.1364/AO.43.003489
- XIONG, X., J. SUN, X. XIE, W. L. BARNES, AND V. V. SALOMONSON. 2010. On-orbit calibration and performance of Aqua MODIS reflective solar bands. *IEEE Trans. Geosci. Remote Sens.* **48**: 535–545, doi:10.1109/TGRS.2009.2024307
- YANG, S. L., AND OTHERS. 2008. Spatial and temporal variations in sediment grain size in tidal wetlands, Yangtze Delta: On the role of physical and biotic controls. *Estuar. Coast Shelf Sci.* **77**: 657–671, doi:10.1016/j.ecss.2007.10.024
- ZHANG, M., J. TANG, Q. DONG, Q. SONG, AND J. DING. 2010. Retrieval of total suspended matter concentration in the Yellow and East China Seas from MODIS imagery. *Remote Sens. Environ.* **114**: 392–403, doi:10.1016/j.rse.2009.09.016
- ZHANG, W. G., X. YU, L. Z. YU, H. FENG, AND M. LIU. 2008. Distinguishing sediments from the Yangtze and Yellow Rivers, China: A mineral magnetic approach. *The Holocene* **18**: 1139–1145, doi:10.1177/0959683608095582
- ZHANG, Y., X. LIU, Y. YIN, M. ZHANG, AND B. QIN. 2012. A simple optical model to estimate diffuse attenuation coefficient of photosynthetically active radiation in an extremely turbid lake from surface reflectance. *Opt. Express* **20**: 20482–20493, doi:10.1364/OE.20.020482

Associate editor: Dariusz Stramski

Received: 17 April 2013

Accepted: 03 September 2013

Amended: 06 November 2013